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Research Article

Estimating leaf area index using unmanned aerial vehicle data: shallow vs. deep machine learning algorithms

Shuaibing Liu,[1,2] Xiuliang Jin ,[1,*,†] Chenwei Nie,[1] Siyu Wang ,[1] Xun Yu,[1] Minghan Cheng,[1] Mingchao Shao,[1] Zixu Wang ,[1] Nuremanguli Tuohuti,[1] Yi Bai [1] and Yadong Liu[1]

1 Institute of Crop Science, Chinese Academy of Agricultural Sciences, Beijing 100097, China
2 School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China

*Author for communication: jinxiuliang@caas.cn
†Senior author

X.J., C.N., and S.L. were involved in study design and research. S.L., S.W., X.Y., M.C., M.S., Z.W., N.T., Y.B., and Y.L. performed the research. S.L. contributed new analytic/computational tools. S.L. analyzed the data and wrote the paper.
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Abstract

Measuring leaf area index (LAI) is essential for evaluating crop growth and estimating yield, thereby facilitating high-throughput phenotyping of maize (*Zea mays*). LAI estimation models use multi-source data from unmanned aerial vehicles (UAVs), but using multimodal data to estimate maize LAI, and the effect of tassels and soil background, remain understudied. Our research aims to (1) determine how multimodal data contribute to LAI and propose a framework for estimating LAI based on remote-sensing data, (2) evaluate the robustness and adaptability of an LAI estimation model that uses multimodal data fusion and deep neural networks (DNNs) in single- and whole growth stages, and (3) explore how soil background and maize tasseling affect LAI estimation. To construct multimodal datasets, our UAV collected red–green–blue, multispectral, and thermal infrared images. We then developed partial least square regression (PLSR), support vector regression, and random forest regression models to estimate LAI. We also developed a deep learning model with three hidden layers. This multimodal data structure accurately estimated maize LAI. The DNN model provided the best estimate (coefficient of determination [R^2] = 0.89, relative root mean square error [rRMSE] = 12.92%) for a single growth period, and the PLSR model provided the best estimate (R^2 = 0.70, rRMSE = 12.78%) for a whole growth period. Tassels reduced the accuracy of LAI estimation, but the soil background provided additional image feature information, improving accuracy. These results indicate that multimodal data fusion using low-cost UAVs and DNNs can accurately and reliably estimate LAI for crops, which is valuable for high-throughput phenotyping and high-spatial precision farmland management.

Introduction

The leaf area index (LAI) is typically defined as the leaf area per unit ground area (Ross, 2012), is an important component of crop phenotypic parameters (Singh et al., 2017; Watanabe et al., 2017), and forms a substantive basis for evaluating crop growth status and for forecasting production (Gholz, 1982). The LAI is also related to numerous physiological processes, such as photosynthesis and canopy light radiation transmission (Landsberg and Waring, 1997; Arias et al., 2007).

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At present, ground observation methods for determining the LAI can be divided into indirect measurement and direct measurement methods (Gower et al., 1999). Direct measurement of the LAI refers to the estimated leaf surface area collected through direct observation in the field or laboratory. Sample collection is more suitable for ecosystems with relatively short plants, such as grassland, tundra, and farmland (Su et al., 2012). Since direct measurement methods are relatively time-consuming and destructive and cannot provide continuous measurements, crop experiments under field conditions typically use indirect observation to estimate LAI data. The indirect measurement method determines the LAI via an optical measurement, which reduces damage to the vegetation and environment, and is faster and more convenient than the direct measurement method (Fang et al., 2019; Yan et al., 2019).

Depending on the optical measurement principle, the indirect method may be based on a radiation measurement (Iiames et al., 2020) or an image measurement (Solberg et al., 2006; Juarez et al., 2009; Brown et al., 2020). Since destructive measurements can only collect crop information once from a given maize (*Zea mays*) plot, they cannot be used for long-term monitoring of crop development (Potithep et al., 2013). Therefore, research into the LAI of maize often uses indirect methods, as exemplified by the Licor LAI-2000 plant canopy analyzer (de Mattos et al., 2020), the Sunscan canopy analyzer (Wilhelm et al., 2000), and so on.

Unmanned aerial vehicles (UAVs) equipped with different micro-sensors can be used to construct a field phenotype observation platform to quickly, nondestructively, and without contact collect high-throughput, ultralow altitude remote-sensing data (Colomina and Molina, 2014).

Although the satellite remote sensing platform can obtain the crop image data in a large area, it is difficult to obtain accurate physiological and biochemical information of crop canopy from the satellite images with low spatial and temporal resolution (Matese et al., 2015). Compared with the satellite remote sensing monitoring platform, UAVs have become a useful platform for monitoring field crop phenotyping information because of their simple operation, flexible flight patterns, and low cost (Yang et al., 2017). In an example of using UAVs to collect phenotype data, Kanning et al. (2018) used an airborne hyperspectral scanner to obtain field images of wheat (*Triticum aestivum*) under nitrogen stress and constructed a multiple linear regression (MLR) model to estimate the LAI of wheat (coefficient of determination [R^2] = 0.79, root mean square error [RMSE] = 0.182). In another work, Duan et al. (2019) estimated the LAI of rice (*Oryza sativa*) based on spectral and texture features and used a Fourier spectral analysis of texture based on UAV RGB images to improve estimation of the LAI of rice. Li et al. (2019) extracted the vegetation index and texture information to construct a normalized-differential texture index and estimated the LAI by using MLR and the random forest (RF) model. Yue et al. (2018) used the vegetation index weighting method to estimate the LAI from UAV RGB and hyperspectral images and obtained $R^2 = 0.59$ with RF regression (RFR). Although these studies propose methods for estimating the LAI of crops using various remote-sensing platforms, they do not consider how the soil background in the images and the different growth stages (e.g. the tasseling stage) affect the LAI.

In previous studies, the spectral, structural, thermal, and textural information, which related to various plant physiological and biochemical parameters, extracted from multi-source UAV data has been used for phenotypic studies of various plants. For example, information on the canopy spectrum obtained from UAV-based hyperspectral and multi-spectral images was used to accurately estimate crop physiology (Houborg and Boegh, 2008), such as the LAI (Qi et al., 2020), nitrogen concentration (Zheng et al., 2018b), leaf chlorophyll content (Xu et al., 2019), and crop yield (Zhou et al., 2017b). Canopy texture based on UAV images has also been used to estimate plant characteristics such as crop biomass (Hlatshwayo et al., 2019; Zheng et al., 2019), LAI (Duan et al., 2019; Li et al., 2019), chlorophyll content (Qiao et al., 2020), and nitrogen concentration (Zheng et al., 2018a). Most of this research has used single sensors, which means that the data structure is relatively simple. The strategy of coordinated observation with multiple remote sensors to estimate crop parameters via data fusion has yet to be fully explored. In previous studies, the fusion of data derived from multiple sensors mounted on UAVs has shown promise for use in estimating crop yield and plant traits (Maimaitijiang et al., 2017).

In recent years, the continuous development of computer hardware has facilitated the progress of machine learning, which has become an active research area in agricultural quantitative remote sensing (Kamilaris and Prenafeta-Boldu, 2018). In particular, deep neural networks (DNNs) have produced good results in machine learning, especially in classification problems. DNN models can be generated after learning training samples and accurately estimating location samples (Jin et al., 2020). Apolo-Apolo et al. (2020) established a ground-based high-throughput phenotyping platform to obtain wheat RGB images and set up a multilayer perceptron model to estimate the LAI. Houborg and McCabe (2018) proposed a hybrid training method to estimate the LAI. The vegetation index, constructed from Landsat 8 remote-sensing satellite images, was used for cubic regression and RFR to estimate the LAI and produced $R^2 = 0.90$. At the same time, other methods based on statistics and machine learning regression, artificial neural networks (ANNs; Yuan et al., 2017), MLR (Jin et al., 2017), support vector regression (SVR; Kuwata and Shibasaki, 2016), RFR (Aghighi et al., 2018), and partial least square regression (PLSR; Rischbeck et al., 2016) have been shown to produce accurate estimates of the yield of various crops. However, few studies have used deep learning (DL) to estimate the LAI of crops, and none have used fused data from various UAV platforms within the framework of DL to estimate the LAI of crops (Jin et al., 2020).

Taking all this into consideration, the present work uses multi-source remote-sensing data, including RGB images, multispectral (MS) images, and thermal infrared (TIR) images, collected from a UAV crop high-throughput phenotyping platform, to develop a multimodal data-processing framework to estimate the LAI of maize. The proposed DL framework is based on DNN architecture, and the results are compared with SVR, RFR, and PLSR machine learning regression algorithms. This work makes several contributions: (1) it proposes a framework for processing fused, multi-source remote-sensing data, which produces multimodal data sets to improve estimates of the LAI; (2) it evaluates the robustness and adaptability of the model for estimating the LAI in single- and whole-growth stages using multimodal data fusion and DNNs; and (3) finally, it explores how field variables, such as nitrogen gradient, lodging, soil background, and maize tasseling, affect the model for estimating the LAI.

Results

LAI estimation for optimal single growth period

To establish an LAI estimation model, the usual practice was to use the LAI at the bottom of the canopy observation value as the ground truth. This study thus estimates the LAI from the bottom of the maize canopy. In this study, the UAV multi-source data (RGB + MS + TIR) were combined as multimodal data to explore the ability to estimating maize LAI. The data from six growth stages of maize were analyzed, and the best estimates for a single growth period were selected as the result (see Figure 1). The estimation accuracy of the machine learning model for LAI in the optimal growth period (8 August) is acceptable. Compared with other observation periods, the average R^2 of the LAI estimation model was 0.2 greater and the relative RMSE was 13% greater than the LAI results in other observation periods. The DNN model provided the best accuracy, with $R^2 = 0.89$, RMSE = 0.47, and relative RMSE (rRMSE) = 12.92%.

Table 1 lists the LAI estimated by various models for the optimal growth period. The best machine learning model is the three-layer DNN model. The model verification gives $R^2 = 0.89$, RMSE = 0.47, and rRMSE = 12.92%. The multimodal data-processing frame, which integrates the maize information from different images, helps the machine learning model learn the characteristic information of the maize LAI.

Machine learning models excel at discovering connections between data. However, for a small sample (72), the data sample distribution is insufficient, resulting in weak generalization (robustness) of the model for the single growth stage. In general, all models for estimating the LAI produce better results from multi-modal data (RGB + MS + TIR) (Table 1). Compared with other machine learning models, the DNN model learns more features from small samples and thus produces better results. At the same time, the DNN model adjusts the depth of the network and the number of nodes to prevent overfitting the estimation model. Based on the results, the image resolution may also affect the LAI estimate. With a single data source, the accuracy of RGB, MS, and TIR follows the same decreasing trend as the image resolution, and the accuracy of TIR + RGB is better than that of TIR + MS. This work uses a sliding 7×7 window to extract the gray-level co-occurrence matrix (GLCM) information from the image. The image resolution thus strongly affects its computational efficiency and accuracy, which may explain why the accuracy follows a trend that is consistent with the resolution. At the same time, the TIR effect also may be used to estimate maize LAI. For example, the combination of RGB + TIR produces a much more accurate estimate than a single RGB data source. However, MS + TIR is less accurate than MS alone. The resolution of MS and TIR images is less than that of RGB images. The extracted TIR texture information is not sufficiently accurate, so the MS + TIR result is less accurate than the MS result.

LAI estimation for whole growth period

The LAI observation data for all growth stages served as a new dataset to evaluate the robustness of the LAI estimation model (Table 2). To estimate the LAI of the whole maize growth period, the DNN, PLSR, support vector machine (SVM), and RFR models have constructed again with the same model structure and parameters as used for the single growth period. The results for the verification of the PLSR model over the whole maize growth period are $R^2 = 0.70$, RMSE = 0.49, rRMSE = 12.78%. Figure 2 shows the reliable accuracy of the maize LAI estimation model constructed from multimodal data.

Spectral saturation appeared in the later stage of maize. The LAI increases with maize growth, but the spectral index extracted from the images remains constant. This leads to an underestimation of the LAI, as shown in the dashed blue circles in Figure 2. Compared with the single growth period, the PLSR model produces the best estimate over the whole growth period.

Stepwise selection of feature information for training

During the segmentation of the vegetation index threshold, the maize leaves appearing in the UAV image produce areas of high reflectivity. This may be because the solar zenith angle is large and the light is sufficient, which generates image noise (Poblete-Echeverria et al., 2017). When extracting canopy vegetation coverage, some of the pixels imaging the high-reflectivity signal from the leaves are, due to their high DN, incorrectly identified as soil background and thus segmented by the threshold after the vegetation index is calculated. The irrigation of different maize germplasm plots in the field also affects image recognition. Given the uneven terrain, the soil color of the low-lying land becomes darker. The vegetation index calculated from the lower layer of the leaves approaches the calculated value of the soil. Although the wind causes some blurred areas in the UAV image, the good stability and superior performance of the UAV RGB

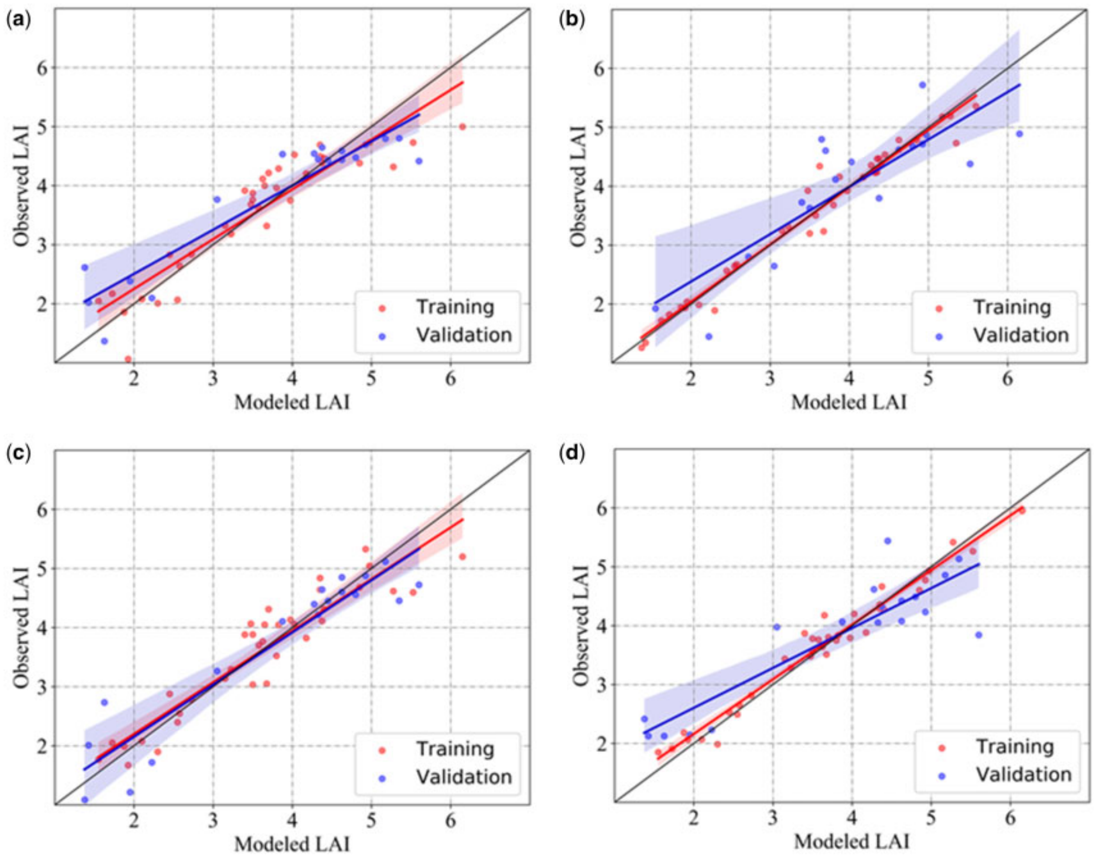

Figure 1 Training (48 samples) and validation (24 samples) of LAI estimation with machine learning model in single growth stage by using multimodal data (8 August). Note: Training and validation of the LAI estimation model: (A) DNN; (B) PLSR; (C) SVR; (D) RFR. Red and blue lines are data fitting lines, black line is 1:1 line, the color strip is 95% confidence interval of the sample.

images make it an important data source for field canopy coverage (CC; Ballester et al., 2017; Zhou et al., 2017b).

The maize CC, texture information, and vegetation index are extracted from the RGB image. The vegetation index and texture information of the red edge and the near-infrared band are constructed from the MS image. The canopy temperature information and texture information are extracted from the TIR image after temperature calibration. The correlation coefficient (Figure 3) is constructed from the LAI and the image feature obtained from the UAV remote-sensing data. To facilitate data presentation, we only display the correlation analysis of the image variables and the LAI of the optimal single growth period (August 8, 2020). The correlation coefficient graph shows that the vegetation index constructed from the visible band is more strongly correlated with the LAI than the vegetation index constructed from the near-infrared and red-edge bands. The GLCM feature obtained from the TIR band also shows potential for evaluating the LAI of the maize canopy. The results show that the maize canopy thermal information obtained from the TIR band is strongly correlated with the LAI. This means that TIR sensors, which are often used in the field in the context of a vegetation evapotranspiration model, are also quite effective for estimating the LAI.

We gathered the data to estimate the LAI, with the results shown in Figure 4. As the parameters increase, the error in the LAI estimate decreases. For a single data source, the estimates of the LAI from the model constructed in this study differ clearly (Figure 4, A and B). Of all the single-modal modeling results, RGB images provide the best estimates of the LAI. Compared with MS and TIR images, RGB images provide better ground resolution. In addition, the morphological and texture information of maize is accurately displayed on RGB images, which allows the machine model to learn most of the characteristics of the maize LAI. In the estimation model constructed from the two data sources, RGB and TIR images produce the best LAI estimates. Compared with MS images, RGB images provide higher spatial resolution, which makes the RGB + TIR estimate more accurate than that of MS + TIR.

DNN models produce better results from multimodal data (RGB + MS + TIR; see Figure 4) than from single- or dual-mode data in whole maize periods. Some texture information obtained from the GLCM is also highly correlated with the LAI. From the remote-sensing images, the gray co-occurrence matrix extracts information

Table 1 Training and validation statistics for various LAI estimation models for maize

Sensor Type	Feature Type	Features	Metrics	DNN		PLSR		SVR		RFR	
				Training	Validation	Training	Validation	Training	Validation	Training	Validation
RGB	CC	36	R^2	0.81	0.82	0.80	0.86	0.86	0.67	0.96	0.71
	VI		RMSE	0.48	0.59	0.50	0.52	0.45	0.65	0.24	0.67
	GLC		MAE	0.38	0.47	0.41	0.44	0.34	0.53	0.18	0.56
	M		rRMSE (%)	13.21	16.03	13.59	14.23	12.37	17.87	6.49	18.41
MS	VI	41	R^2	0.82	0.70	0.81	0.76	0.86	0.57	0.93	0.68
	GLC		RMSE	0.47	0.77	0.48	0.68	0.45	0.74	0.31	0.72
	M		MAE	0.36	0.6	0.37	0.56	0.36	0.61	0.25	0.59
			rRMSE (%)	12.86	21.04	13.20	18.64	12.36	20.31	8.35	19.60
TIR	°C	4	R^2	0.61	0.51	0.65	0.56	0.58	0.55	0.89	0.54
	GLC		RMSE	0.68	0.98	0.65	0.93	0.77	0.84	0.39	0.85
	M		MAE	0.54	0.78	0.51	0.75	0.54	0.67	0.30	0.68
			rRMSE (%)	18.71	26.81	17.83	25.44	21.16	23.02	10.57	23.25
RGB + MS	CC	77	R^2	0.86	0.69	0.83	0.87	0.96	0.62	0.97	0.81
	VI		RMSE	0.41	0.77	0.45	0.51	0.25	0.70	0.20	0.55
	GLC		MAE	0.32	0.62	0.36	0.4	0.19	0.59	0.16	0.43
	M		rRMSE (%)	11.10	21.15	12.61	13.93	6.97	19.25	5.39	15.15
RGB + TIR	CC	40	R^2	0.82	0.88	0.82	0.84	0.88	0.72	0.97	0.70
	VI		RMSE	0.47	0.49	0.46	0.55	0.42	0.61	0.22	0.69
	GLC		MAE	0.38	0.40	0.37	0.44	0.33	0.52	0.16	0.58
	M		rRMSE (%)	12.86	13.31	12.70	15.07	11.57	16.59	5.90	18.78
	°C										
MS + TIR	VI	45	R^2	0.79	0.62	0.82	0.73	0.89	0.56	0.93	0.75
	GLC		RMSE	0.50	0.86	0.47	0.73	0.41	0.75	0.32	0.63
	M		MAE	0.41	0.68	0.38	0.60	0.33	0.62	0.25	0.51
	°C		rRMSE (%)	13.74	23.46	12.76	19.94	11.14	20.55	8.82	17.37
RGB + MS + TIR	CC	91	R^2	0.87	0.89	0.84	0.85	0.96	0.69	0.97	0.76
	VI		RMSE	0.40	0.47	0.44	0.54	0.23	0.64	0.20	0.62
	GLC		MAE	0.32	0.38	0.35	0.45	0.19	0.54	0.16	0.49
	M		rRMSE (%)	10.91	12.92	11.95	14.86	6.32	17.48	5.50	16.85
	°C										

Table 2 Estimates of LAI over whole growth period

Metrics	DNN		PLSR		SVR		RFR	
	Training	Validation	Training	Validation	Training	Validation	Training	Validation
R^2	0.80	0.66	0.78	0.70	0.85	0.69	0.95	0.59
RMSE	0.42	0.52	0.44	0.49	0.37	0.50	0.21	0.58
MAE	0.34	0.41	0.35	0.41	0.28	0.42	0.16	0.46
rRMSE (%)	10.98	13.61	11.51	12.78	9.60	13.08	5.43	15.00

sensitive to the maize LAI (Zhou et al., 2017a). Vegetation-structure information related to leaf width and leaf length is related to the LAI and to photosynthetically active radiation. The parameters, such as leaf width and leaf length, in leaf structure, can provide information for inverting the LAI (Shi et al., 2021).

The Spearman's rank correlation coefficients between feature parameters and the LAI (Figure 5) show that a considerable number of image variables are highly correlated with the observations of the second and third layers of the LAI. Maize leaves have the form of narrow strips, so leaves in the top layer cover those in the bottom layer more than those in the middle layer. Therefore, some image feature variables have higher LAI correlation coefficients for the second and third layers. The UAV phenotyping platform mainly collects data via vertical or oblique observation, so the observation penetrates the canopy and detects the leaves under the canopy. This also improves the ability of image feature variables to estimate the LAI of the second and third leaf layers. The orthophotos obtained by the UAV also reflect the LAI distribution in the middle layer of maize.

A step regression model is constructed to analyze how the feature information affects the estimate of the LAI. The forward stepwise regression results are shown in Figure 6. The model was built from a different number of parameters that traverses all feature information each time. Better results are obtained for the soil adjusted vegetation index (SAVI) and visible atmospherically resistance index (VARI) than is possible with a variety of models. The correction of soil and atmosphere may explain their high weight in the model. Soil pixel information in the field accounts for a large fraction of the community, and SAVI can eliminate the impact of soil pixel reflectivity on the LAI estimate. At the same time, the different weather while collecting the field data means that the correction of the atmospheric environment plays the role of the VARI.

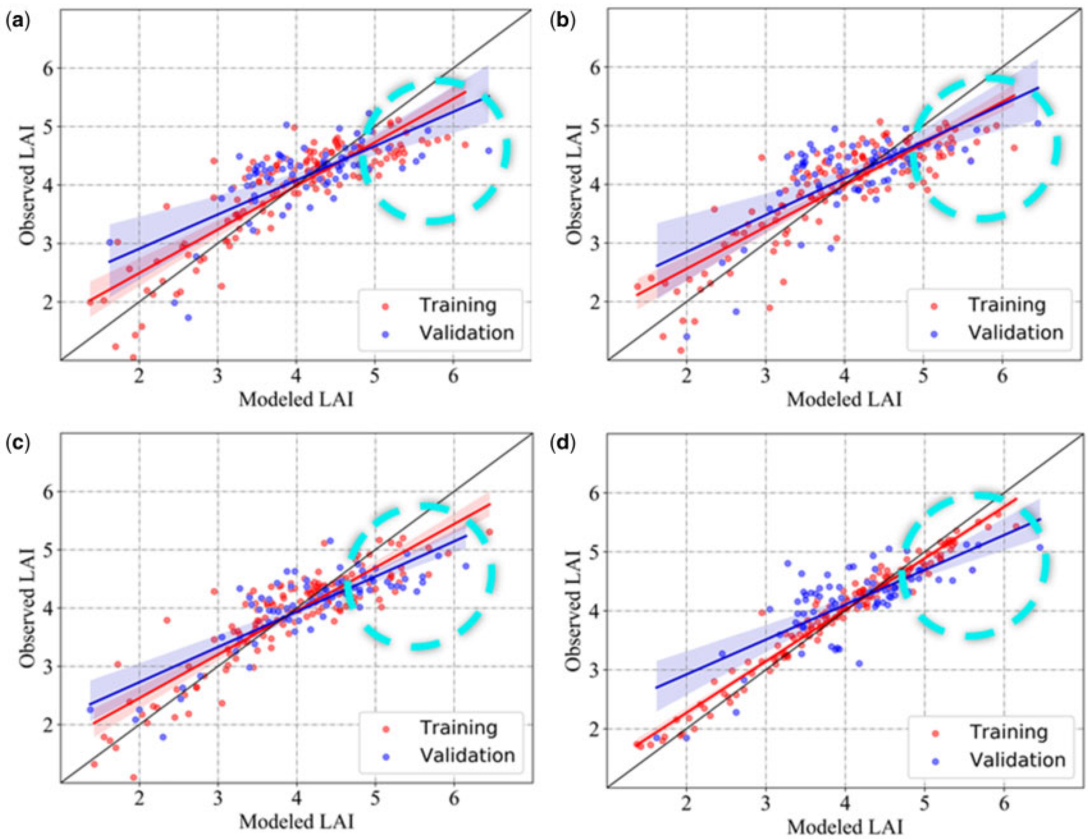

Figure 2 Training (301 samples) and validation (150 samples) of LAI estimates with machine learning model over whole growth stage. Note: (A) DNN; (B) PLSR; (C) SVR; (D) RFR. Red and blue lines are data fitting lines, black line is 1:1 line, the color strip is 95% confidence interval of the sample, and the blue dotted line represents the LAI underestimation area.

Influence of soil background and maize tassels

Figure 7 shows the results of vegetation coverage threshold segmentation in maize. The threshold segmentation of the vegetation index on August 8, 2020 is calculated by using the Python-GDAL 2.2.4 image-processing database. After removing the soil background, we reconstructed the estimation model to allow us to discuss how the soil pixels affect the maize LAI. The correlation coefficient between vegetation CC and the LAI is used to evaluate the accuracy of the six classification models because the actual field CC is difficult to observe as the ground truth. The soil segmentation results most strongly correlated with the maize LAI are used to extract the image feature information. Table 3 lists the extraction results.

As can be seen in Table 3, the RGBVI is more correlated with the LAI than are the other vegetation indexes constructed from the threshold segmentation models. The results of the threshold segmentation mask map show that VEG (Figure 7A), VDVI (Figure 7B), and EXGR (Figure 7D) are not effective for classifying maize. For extracting the maize seedling information, most of the canopy texture information is lost. The indexes RGBVI (Figure 7C), CIVE (Figure 7E), and EXG (Figure 7F) can effectively extract the canopy information of maize seedlings and retain the relatively complete maize morphology, texture, and other important features. However, EXG and CIVE are not sufficiently accurate to recognize the edge information of maize seedlings, and instead produce redundant edge information.

The lack of recognition of the bottom leaves results in substantial loss of canopy information, which may also explain the low correlation between EXG, CIVE, and the LAI. The RGBVI retains more useful information for maize canopy plots that have grown well, and the accuracy with which canopy information is extracted at the seedling stage can be very reliable. Therefore, we chose the CC classification extracted by using the RGBVI as the canopy structure extraction information.

The CC extracted previously is used as a quantitative index to construct the model data set. In this study, the soil and other ground feature information were removed from the UAV images to analyze the contribution of soil pixels to the LAI. This quantitative information was re-extracted from the remaining maize pixels. Figures 8, A and 10, C showed that, under different light conditions and soil irrigation conditions, the vegetation canopy can be extracted from the image by exploiting the vegetation index threshold. We use data from the whole growth period as training data, separate the images into original images and images without soil (Figures 8, D and 10, F), and build the machine learning model to estimate the maize LAI.

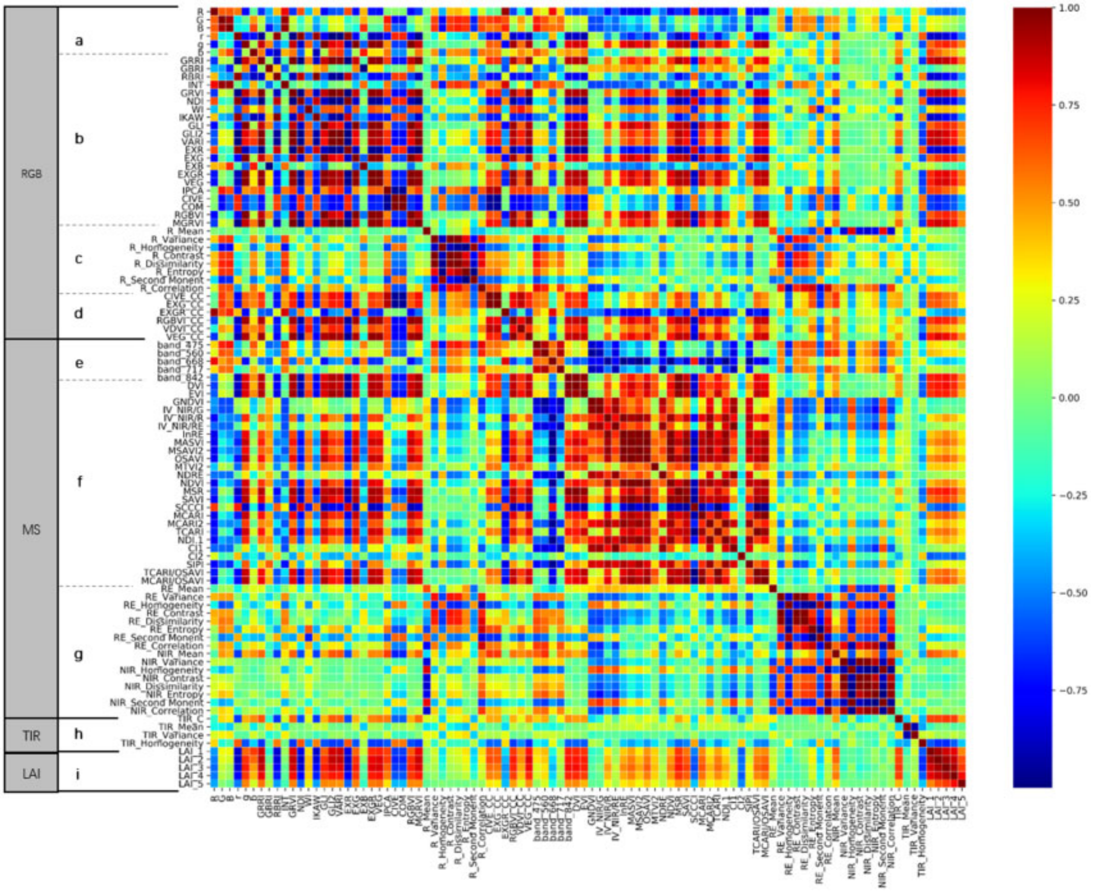

Figure 3 RGB, MS, and TIR image features, and map of LAI spearman correlation coefficient (scale to right). (A) RGB-original band, (B) RGB-VI, (C) RGB-GLCM, (D) RGB-CC, (E) MS-original band, (F) MS-VI, (G) MS-GLCM, (H) TIR-temperature and GLCM, (I) LAI-vertical distribution.

Upon removing the soil background from an image, the estimation accuracy generally decreases and rRMSE increases significantly (see Table 4). The maize leaves in the lower layer are mistakenly identified as the soil background and removed, which means that part of the spectral information on the canopy leaves is lost. After removing all the soil background, we only collect the pixel information of the leaves, which saturates the spectral information and reduces the model accuracy. When using the Sunscan instrument to collect data involving photosynthetically active radiation, some of the solar radiation obtained comes from sunlight reflected by the soil or from secondary refraction of the leaves. After filtering out the information from the soil pixels and the underlying leaves, this part of the image feature is lost, but the total canopy multimodal data are still used for modeling, which may also contribute to the estimation error.

Previous studies show that maize tassels contribute to the canopy spectral information (Stewart et al., 2003). Figure 9 shows that, in the vertical photography mode of the UAV, the tassel area on the image increases during the later stages of maize growth. The results allow us to discuss the response of the tassel to the canopy spectrum after tasseling.

Table 5 shows that rRMSE of the LAI estimation model constructed using no-tasseling images improves significantly. During the maize tasseling stage, the tassel area increases in the image and obstructs the lower leaves, causing loss of leaf information. At the same time, because the dataset was classified according to the growth period of maize, the color and texture information of maize leaves also differed before and after tasseling. This may also decrease the estimation accuracy from models based on UAV images after maize tasseling. However, the contribution of tassels to the canopy radiant energy transmission and the response of the canopy spectrum remains to be studied.

Discussion

Contribution of multimodal data to estimation of LAI

For the LAI data of the whole growth period, we did not change the structure of the DNN model, which, depending on the distribution of the sample datasets, may improve the accuracy of LAI estimates. It would thus be desirable in follow-up research to further explore the applicability of the DL models.

At the same time, the field canopy temperature data extracted by the TIR sensor, which is usually sensitive to crop parameters such as water content, pigment concentration, and characteristics of canopy structure, is helpful for estimating the vegetation canopy LAI (da Luz and Crowley, 2010; Elarab et al., 2015; Neinavaz et al., 2016). Previous

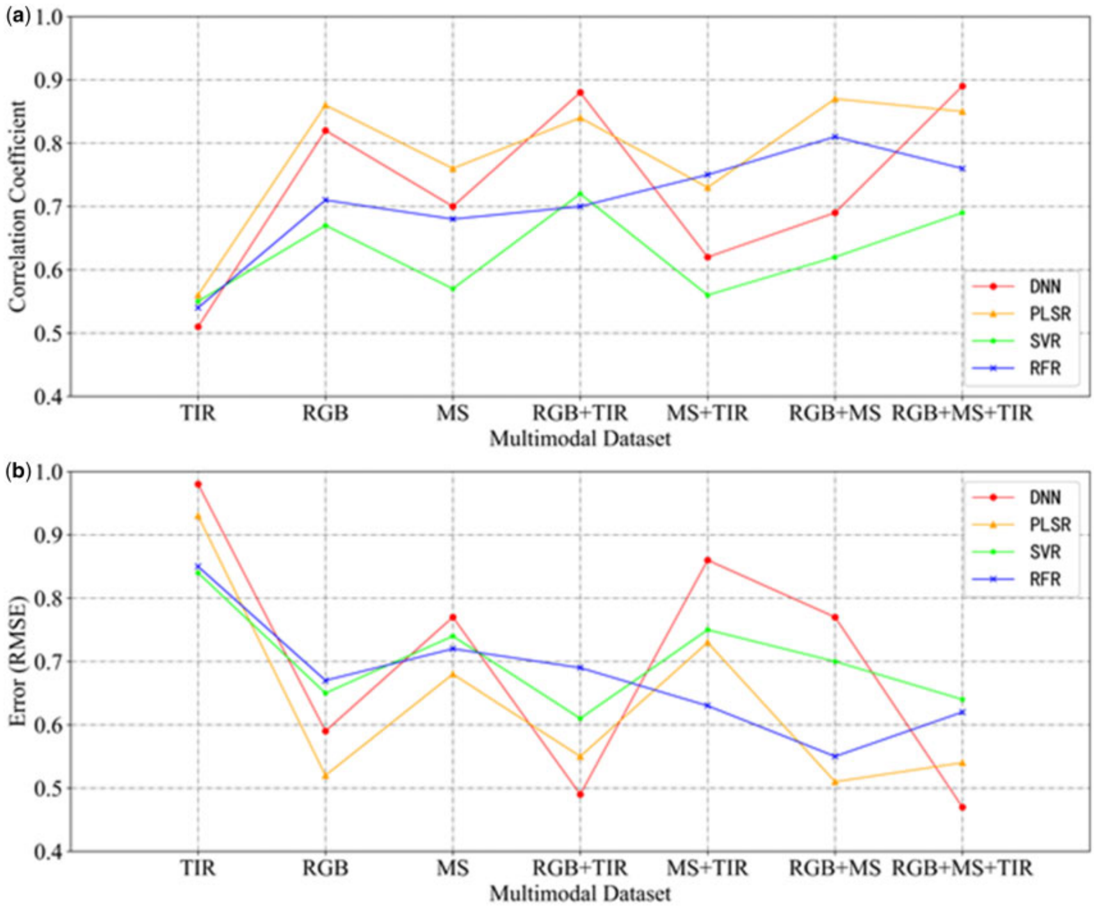

Figure 4 Maize LAI estimation performance of different models with various input feature types and numbers. Note: (A), (B): R^2 and *RMSE* different input features from multimodal data sources (RGB, MS, and TIR).

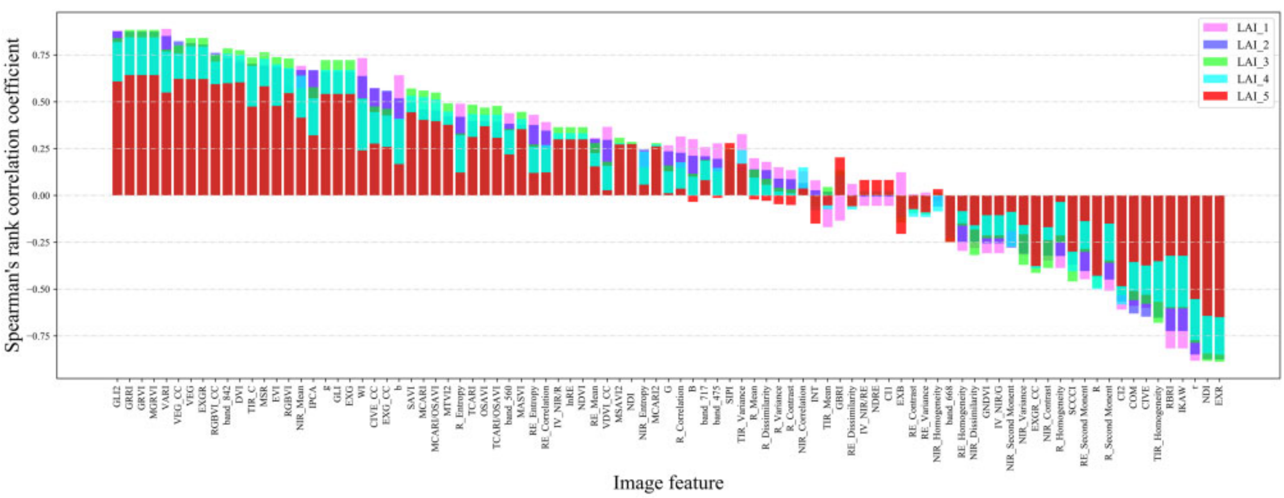

Figure 5 Spearman's rank correlation coefficient between the image feature and the maize LAI.

studies have confirmed that coupling spectra with the characteristic variables of other sensors can improve the estimation accuracy of the model (Geipel et al., 2014; Bendig et al., 2015; Stanton et al., 2017). The results of the RGB + TIR multimodal data are like those of the RGB + MS + TIR multimodal data, with high estimation accuracy. When collecting the field-heat data in this study, we calculated the averaged heat of the maize germplasm plot, and this value served as a variable to train the estimation model. The LAI in the field can represent the density of leaves to a certain extent. Places, where leaves are denser and thus collect more radiation, are hotter than places where leaves are sparse, which may explain why thermal information can improve the accuracy of the model. To better understand how canopy heat data is related to the LAI, especially as a function of plant species, development stages, environmental conditions, and other factors and their interactions must be further researched.

The climate disasters of 2020 caused large areas of lodging in the field, and this part of the test plot was not observed in making these LAI models. In the present work, the LAI field observations were only carried out in maize germplasm plots without lodging. Therefore, the data distribution was relatively uniform, which may explain the better fitting effect of the linear model (Varoquaux, 2018). At the same time, the COVID-19 (Corona Virus Disease 2019) pandemic of 2020 prevented access to the field in the early stage of maize growth. When the data were collected, the LAI in the field was already in a saturated state, so no obvious differences appeared in the growth trend of each maize germplasm plot. The distribution of various feature samples is uneven, and most samples are concentrated in a specific interval. For machine learning models, the lack of sufficient training data may also help explain why DNN models are worse than PLSR models over all maize growth stages (Chapelle et al., 2002; Wang and Hebert, 2016).

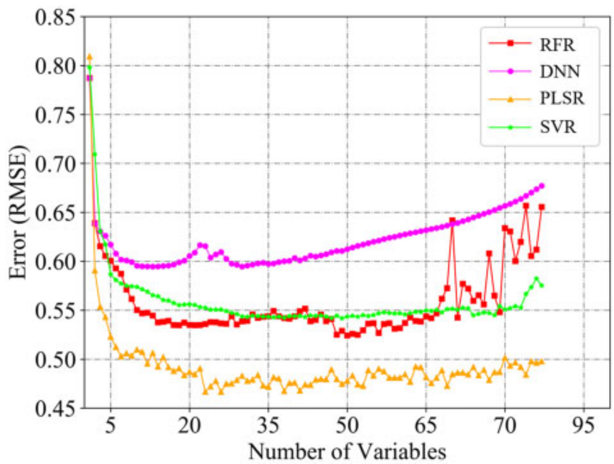

Figure 6 Forward stepwise regression analysis of image features.

Unlike Kanning et al. (2018), who mainly used hyperspectrometer field images to calculate the LAI, this study uses mainly RGB, MS, and TIR cameras to evaluate the LAI, which not only reduces the cost of data collection, but also provides more accurate multimodal estimates of the LAI. For the nitrogen-stress maize germplasm experiment, we also added thermal information to the multimodal data. The RMSE is 0.4 lower than when using an RGB camera to observe a single growth period (Li et al., 2019). At the same time, the GLCM provides the canopy texture information from the MS and TIR images, which was added to the machine learning LAI estimation model. The LAI estimate is more accurate than that obtained by using a single RGB data source (Yue et al., 2018), which shows that the DNN model designed in this study provides reliable accuracy when using multimodal data sources.

Canopy structure information is a promising variable to estimate crop phenotype. Previous studies have also shown that canopy-structure characteristics obtained from UAV images, such as plant height and plant spacing, are good indicators of soybean grain yield given high-throughput phenotypes (Yu, 2016). In this study, the COVID-19 pandemic delayed the acquisition of bare field ground images, so no reliable field surface model could be developed. The height of vegetation is also often used to estimate the LAI (Apolo-Apolo et al., 2020) because plant height is an important phenotypic parameter of vegetation. Therefore, follow-up research should collect data on maize plant height and verify it against the plant height data obtained from the UAV RGB images through structure from motion. It should also be added to the research on crop phenotyping information inversion.

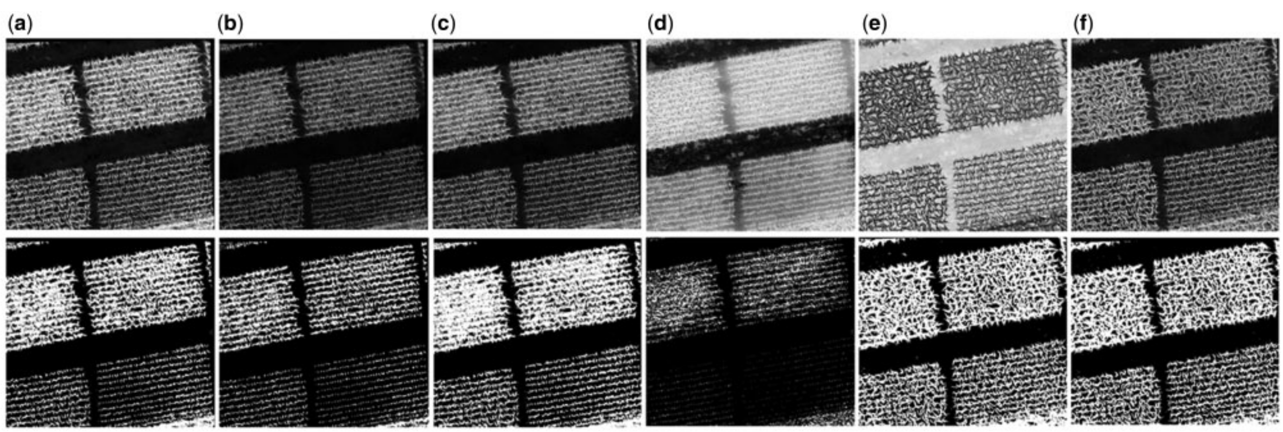

Figure 7 Vegetation index threshold segmentation and binary mask map. The top row of images shows the vegetation index map, and the bottom row of pictures shows the binarized pictures obtained by threshold segmentation, which are (A) VEG, (B) VDVI, (C) RGBVI, (D) EXGR, (E) CIVE, and (F) EXG.

Effects of maize variety and lodging on LAI estimation

The analysis of variance (ANOVA) was set up to analyze how maize breeding germplasm, sowing time, and the two levels of nitrogen stress affected the LAI layered observation. The *F* test and *P*-value in the ANOVA result served as the significant indexes for determining the LAI. The results in Table 6 show that the variety, sowing time, and nitrogen stress gradient all significantly affect the LAI stratification. Figure 10 shows the data distribution of maize LAI.

Table 3 Correlation coefficient between CC and LAI

Vertical Distribution	VEG	VDVI	RGBVI	EXGR	EXG	CIVE
LAI-1	0.38	0.41	**0.60**	0.48	0.09	0.26
LAI-2	0.41	0.44	**0.70**	0.58	0.15	0.29
LAI-3	0.32	0.34	**0.65**	0.54	0.20	0.23
LAI-4	0.27	0.29	**0.53**	0.46	0.17	0.17
LAI-5	0.09	0.10	**0.27**	0.23	0.14	0.06

Note: LAI-1 is the observed LAI from the bottom layer, LAI-5 is the observed LAI from the top layer. The bold type is the better result of vegetation canopy coverage classification.

This study used two maize germplasm materials in two nitrogen-gradient test fields (Figure 11C). Figure 11 shows the vertical distribution of the LAI of the two varieties throughout the growth period (only one variety of JNK-728 is set in the field registration of the N100–N400 gradients). The results show that the LAI evolves in the same way in the two varieties under different nitrogen gradients. The LAI for ZD-958 (Figure 11B) vertically distributed is greater than that for JNK-728 (Figure 11A). Under the nitrogen distributions N0–N10, each ZD-958 layer has a greater LAI and is more adaptable to variations in nitrogen supply. The nitrogen gradient from N100 to N400 more strongly affects the

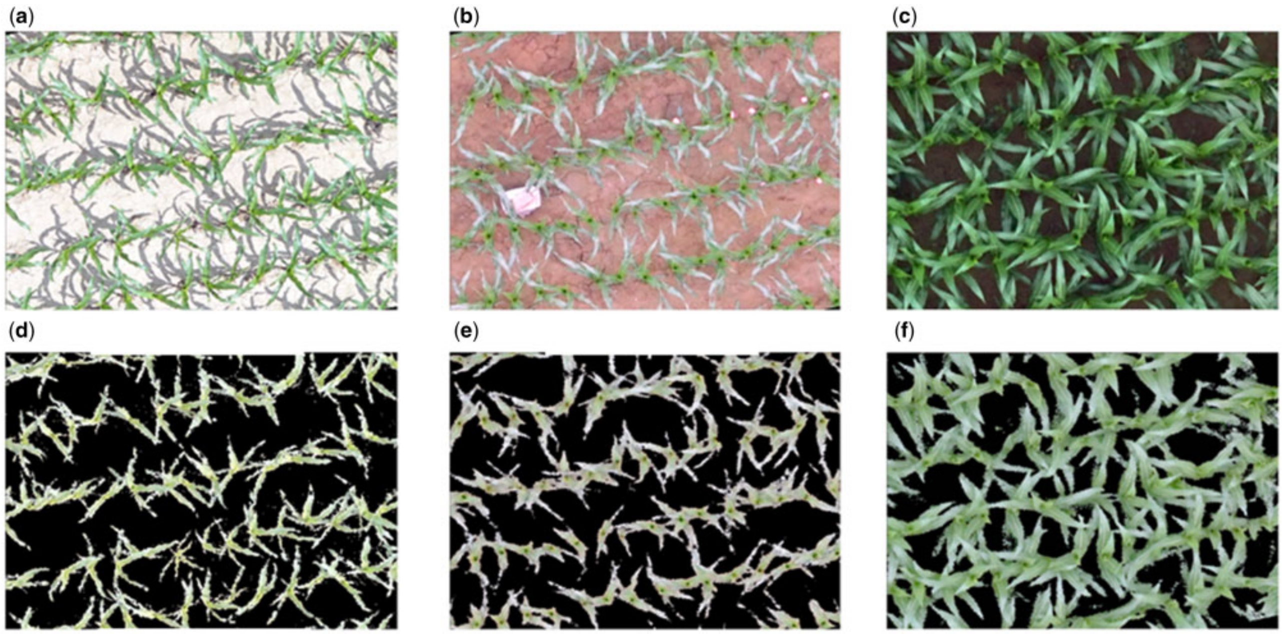

Figure 8 The original image of the maize canopy and images after removing the soil by threshold segmentation. (A–C) Original UAV images from different maize growth periods. (D–F) Images with soil background removed.

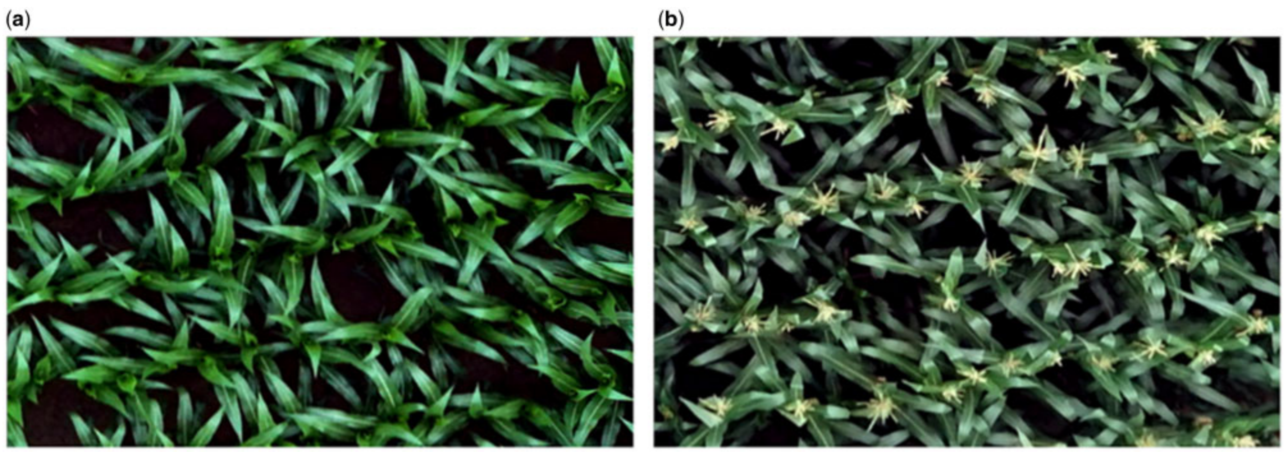

Figure 9 UAV images before and after maize tasseling stage. A, Before maize tasseling. B, After maize tasseling.

Table 4 LAI estimates from machine learning model

Image Type	Metric	DNN		PLSR		SVR		RFR	
		Training	Validation	Training	Validation	Training	Validation	Training	Validation
No Soil	R^2	0.59	0.50	0.56	0.49	0.38	0.13	0.94	0.57
	RMSE	0.56	0.64	0.61	0.60	1.22	1.30	0.22	0.55
	MAE	0.46	0.50	0.45	0.48	0.91	1.03	0.17	0.44
	rRMSE (%)	14.97	16.86	16.22	15.80	24.80	34.11	5.96	14.49
Soil	R^2	0.80	0.66	0.78	0.70	0.85	0.69	0.95	0.59
	RMSE	0.42	0.52	0.44	0.49	0.37	0.50	0.21	0.58
	MAE	0.32	0.42	0.35	0.41	0.27	0.40	0.17	0.45
	rRMSE (%)	10.98	13.61	11.51	12.78	9.60	13.08	5.43	15.00

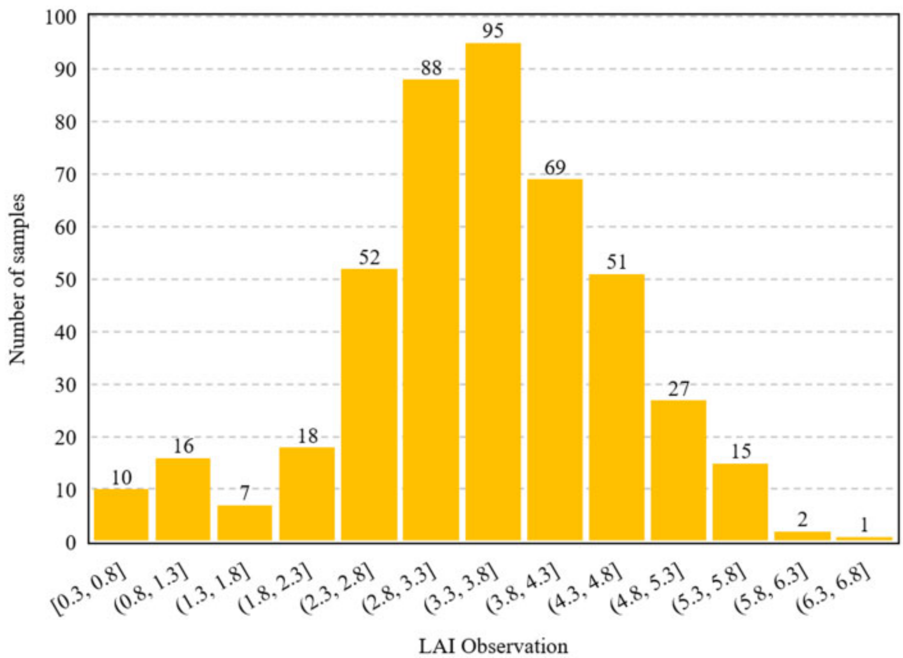

Figure 10 Distribution of LAI acquired over the whole growth period. The number is the number of samples in the LAI interval.

Table 5 LAI estimations by machine learning model

Image Type	Metric	DNN		PLSR		SVR		RFR	
		Training	Validation	Training	Validation	Training	Validation	Training	Validation
No Tassel	R^2	0.58	0.31	0.50	0.33	0.30	0.15	0.88	0.31
	RMSE	0.50	0.57	0.55	0.55	0.83	0.72	0.27	0.56
	MAE	0.40	0.45	0.44	0.48	0.62	0.61	0.21	0.46
	rRMSE (%)	14.54	16.86	16.06	16.09	24.07	20.93	7.93	16.36
Tassel	R^2	0.43	0.36	0.42	0.38	0.03	0.09	0.89	0.34
	RMSE	0.68	0.80	0.69	0.80	1.29	1.25	0.31	0.82
	MAE	0.54	0.64	0.54	0.64	0.89	1.04	0.24	0.66
	rRMSE (%)	17.66	20.79	16.85	20.54	34.99	29.53	7.92	21.24

Table 6 ANOVA result between LAI and field control variables

Control Variable	F	F-crit	*P*-value
Varieties (ZD-958, JNK-728, and FK-139)	21.61	3.02	1.43×10^{-9}
Nitrogen stress (N0–N10)	11.01	3.20	1.28×10^{-4}
Nitrogen stress (N100, N200, N300, and N400)	6.27	2.01	4.26×10^{-6}
Sowing date (20/04, 30/04, 13/05, 23/05, 02/06, 14/06, 24/06, and 04/07)	74.61	2.13	2.05×10^{-32}

LAI than nitrogen application from N0–N10, which increases with increasing nitrogen supply.

The year 2020 saw large-scale wind disasters due to the monsoon climate, which led to large-scale lodging in some plots (see Figure 11D). The red, yellow, and green sections indicate severe, light, and no lodging, respectively. Within a few days after the lodging, bamboo poles were set up to support the maize, but, despite these efforts, the lodging may still affect the observations. Figure 11D shows that plots of N1 and N10 lodging area, compared with other material

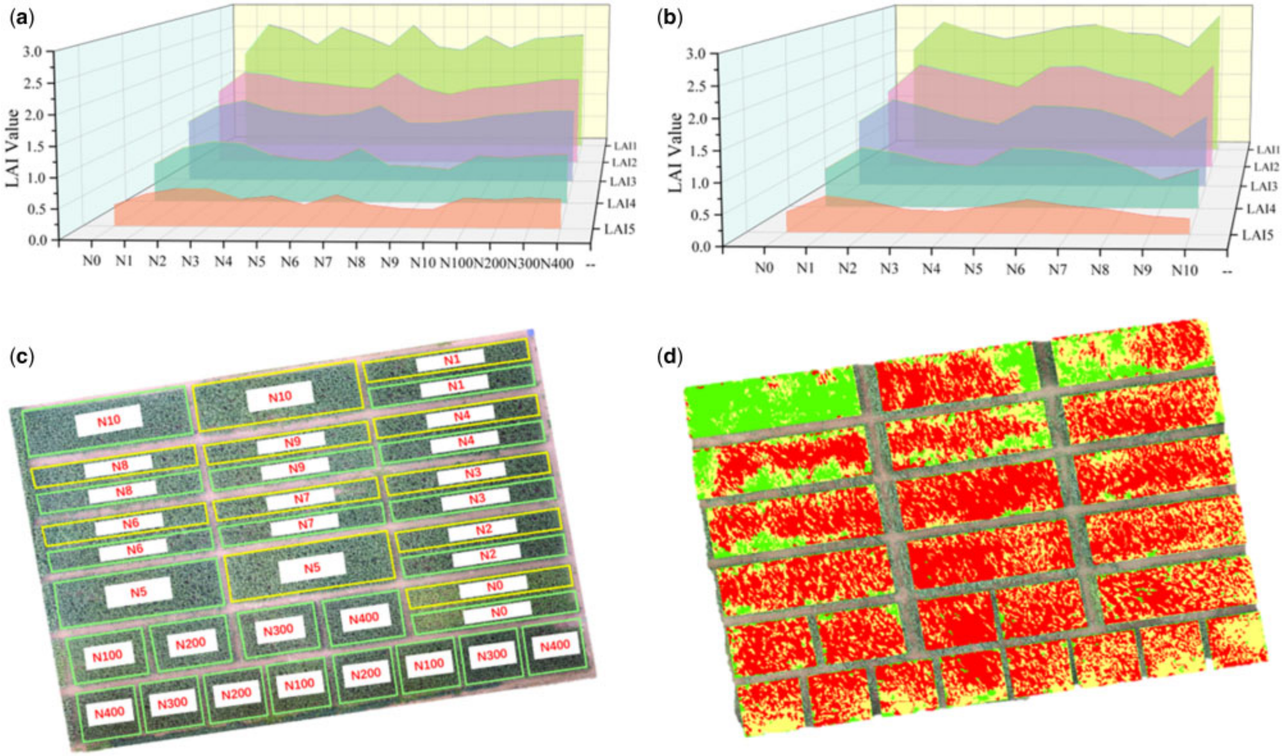

Figure 11 Effects of different nitrogen treatments and maize lodging on vertical distribution of LAI. Note: The vertical distribution of maize LAI for two maize germplasm under two different nitrogen gradients N0–N10 (A) and N100–N400 (B). C, Nitrogen-gradient field experiment, yellow box is ZD-958 and green is JNK-728. D, Maize lodging: red is severe lodging, yellow is slight lodging, green is no lodging.

plots, are smaller. Therefore, the LAI value in plots N1 and N10 exceeds that in other nitrogen experiments.

The proposed DNN model showed good results in terms of accuracy and practicability and can be implemented on the UAV observation platform. Previous studies have rarely discussed how lodging affects the images used for determining the LAI. After lodging, many of the leaves of the maize were broken. Compared with healthy unlodged maize leaves, the effective interception area of sunlight is reduced, which lowers the LAI. At the same time, studies show that the number of healthy unlodged maize plants has an average of 2.2 more leaves than maize plants raised after lodging, and 4.9 more leaves than lodged maize plants that are not raised. The destruction of the maize root system that occurs when maize is raised reduces nutrient absorption and causes some leaves to yellow and wither. This may be another reason for the low LAI of maize.

Future research and enhancements to increase model accuracy

Although the experimental results are encouraging, there is still room for improvement. The proposed DNN model and PLSR model produce good results in terms of accuracy and practicability and can be quickly configured on the UAV observation platform. The sample size of the dataset can be further increased, and the distribution of the data can be increased accordingly. For machine learning models, larger sample numbers improve model training. This is especially true for the DL model, for which a large amount of data with various distribution types are needed to understand the relationship between the data and the estimated value. Because of COVID-19 pandemic in 2020, we lack early UAV image data for maize. At the same time, the early growth stage of maize plants is relatively low. Therefore, the early data of maize only include the data at the lowest level of LAI without vertical distribution. In future experiments, we will analyze how leaf growth conditions at the top and bottom layers affect estimates of the LAI estimation of maize. In this study, the COVID-19 pandemic and extreme weather that caused lodging meant that insufficient data images were collected in the early stage of maize, especially in the first stage of maize growth. At the same time, increasing the field environmental variables or increasing the number of maize germplasm resources would widen the difference in maize growth in the field, which can also increase the abundance of data and lead to more accurate estimates of LAI. Adding biophysical parameters obtained by various sensors, such as the thermal imager in this study, can also improve the accuracy of the model. In addition, sensing devices such as fluorescence observers and RGB-D cameras are also of great substantially for digging deep into these emerging sensing devices to find the corresponds between their characteristics and vegetation phenotypes. Finally, because many environmental factors, and especially weather conditions, affect crop growth, appropriate methods need to be considered in future studies to harness these parameters as quantitative input for model estimation.

The machine learning model is integrated into the Python libraries Keras and Scikit-learn. Based on the DNN model, since the network structure can be flexibly customized, it can accelerate data training and improve model performance (Duveiller, 2014).

This research also indicates that this type of image analysis algorithm can be quickly deployed. These models can be used for rapid phenotype identification in the commercial and breeding fields, so these results are of substantial practical value. At the same time, this work explores the ability of multimodal models to estimate the vertical distribution of LAI. Although the estimated LAI for the bottom of the canopy is accurate, the maize tassels and leaves of varying shapes in the upper maize canopy reduce the accuracy of the estimated LAI in the upper canopy. In addition, the analysis of the vertical distribution of maize LAI is unsatisfactory, and the interpretation of the estimated LAI at different canopy heights is not convincing.

Conclusion

This work constructs a multimodal data processing framework with the help of a UAV crop phenotype observation platform and discusses how environmental factors affect the estimation model. The results show the potential of multimodal data to estimate LAI. The main conclusions are as follows:

(1) We analyze the contribution of different data sources to estimate the LAI and propose a framework for using multi-source remote-sensing data to estimate the LAI. The best estimation machine learning model for a single growth period is DNN ($R^2 = 0.89$, rRMSE = 12.92%), and for the whole growth period, the best model is PLSR ($R^2 = 0.70$, rRMSE = 12.78%).

(2) DNN models produce better results from multimodal data (RGB + MS + TIR) than from single- or dual-modal data, which is more robust than other LAI estimation models. Image resolution is an important factor affecting the accuracy of LAI estimation.

(3) The maize tassels in the upper canopy decrease the accuracy of LAI estimates. The soil background provides additional radiant information for the LAI estimation model and improves the estimation accuracy. In addition, lodging strongly affects the LAI of maize.

These research results show that the use of multimodal data fusion and machine learning has substantial potential for applications. However, to further evaluate the robustness of this method, it should be tested with numerous genotypes under varying crop types, developmental stages, and environmental conditions.

Materials and methods

Experimental site and design

The experimental field for this study was in Xinxiang City, Henan Province, China, which has a warm temperate continental monsoon climate. The average annual temperature and precipitation are 14.1°C and 548.3 mm, respectively. The rainfall is concentrated in July and August. The annual average evaporation, sunshine duration, and frost-free period are 1908.7 mm, 2407.7 h, and 200.5 days, respectively. Three maize germplasm breeding experiments were designed in the field. The first experiment used different sowing dates to explore how sowing date affects the different LAI varieties. The other two maize-breeding experiments used nitrogen stress-gradient treatment with drip irrigation, with a planting density of 75,000 plants/hm^2. Table 7 gives the size of each plot for each sowing date. The nitrogen-gradient experiment used a plot of 9.6 m × 9.0 m (Figure 12B), and the plots for the plant nitrogen experiment were all 26.6 m × 9.0 m (Figure 12C). While collecting the image data of UAV, the LAI observation of maize was collected synchronously in the field. Due to the influence of maize maturity, field irrigation and maize lodging, and other factors, the number of LAI obtained from each image is different. A total of 451 plots of LAI observations were obtained during the whole growth period of maize.

Row planting was used for both the sowing-date and nitrogen-gradient experiments: a plant spacing of 0.23 m width and a row spacing of 0.60 m. The test field used urea (N46%) as nitrogen fertilizer, superphosphate as phosphate fertilizer (P_2O_5, 12%, 90 kg/hm^2), and potassium chloride (K_2O, 60%, 12 kg/hm^2). Phosphorus potassium fertilizer was applied once as a base fertilizer. The irrigation volume of each plot remained constant throughout the experiments. Figure 12 shows the location and layout of the field survey area.

The rational use of nitrogen fertilizer is an effective field management technique to obtain a high yield of maize. Through field experimentation, we analyzed how nitrogen fertilizer affects the LAI of maize. Two strategies were used for the nitrogen fertilizer experiment: The first was to use 11 operational nitrogen gradients, labeled N0–N10, and the second was to use four gradients, labeled N100–N400. Table 8 shows the specific amounts of nitrogen applied. Nitrogen management in the field and applying a nitrogen topdressing during the key growth period of maize plays a vital role in the growth and development of maize.

Data acquisition

Collection of field data

The field experiment used the Sunscan canopy analyzer, which contains a probe (SS1), a sensor (BF5), and a handheld terminal (RPDA2). The data-collection terminal is equipped with Sundata software. The SS1 measures 1,000 × 13 mm^2 and its spectral range is 400–700 nm, and the BF5 measures 120 × 120 × 95 mm^3 and covers a photosynthetically active radiation measuring range of 0–2,500 $\mu mol \cdot m^{-2} \cdot s^{-1}$ (total radiation and scattered radiation). The LAI was collected 5 times. All LAI observations were collected from each layer at four different orientations in the horizontal plane (rotating 45° between each observation). The average value was retained. Analyzing the vertical

Table 7 UAV data collection and flight specification

Date	Growth Stage	Maize Plots Number of LAI Observed	Flight Altitude (m)		Overlap (%)		Spatial Resolution (cm)		
			RGB + TIR	MS	RGB + TIR	MS	RGB	MS	TIR
July 20, 2020	Tasseling	106	30	30	80–70	80–70	0.80	3.98	N/A
August 01, 2020	Silking	72	30	30	80–70	80–70	0.75	2.29	N/A
August 08, 2020	Blister	72	30	30	85–75	85–75	0.99	2.31	4.55
August 16, 2020	Milk	34	30	30	90–80	90–80	0.99	2.23	2.83
August 23, 2020	Dough	88	30	30	90–80	90–80	0.97	2.28	4.78
September 09, 2020	Dent	79	30	30	90–80	90–80	0.99	2.34	8.25

Note: N/A means not available, and no TIR images were collected on July 20 and August 1.

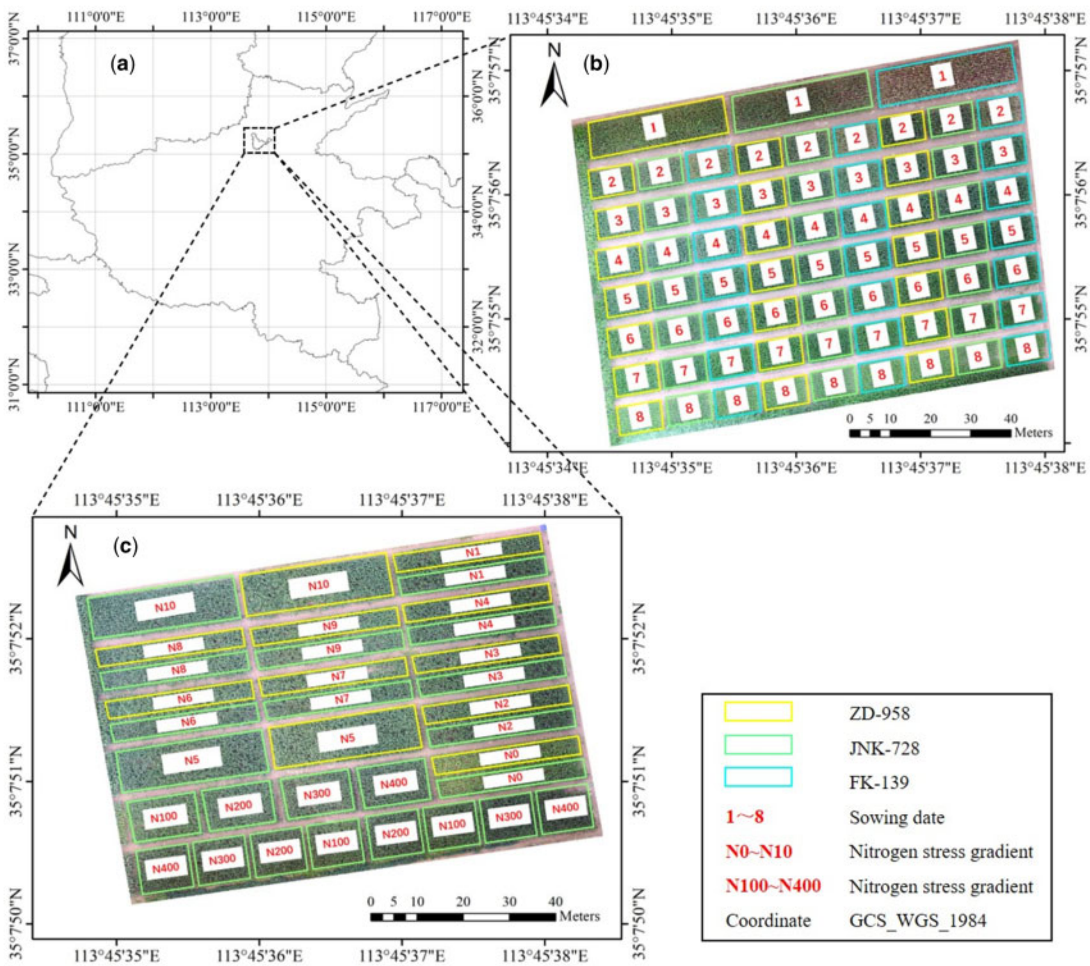

Figure 12 Unmanned Arial Vehicle RGB orthophoto images. Note: A, Experimental maize field, Institute of Crop Science, Chinese Academy of Agricultural Sciences, Xinxiang County, Henan Province, China, (B) in situ LAI sampling locations for different sowing dates for maize germplasm experiment, (C) in situ LAI sampling locations for two experiments involving nitrogen-gradient treatment of maize germplasm, ZD-958, JNK-728, and Fk-139 are three different maize varieties.

distribution of the LAI within the vegetation canopy, estimated from remote-sensing images, can help to determine whether the remote-sensing approach produces more penetrating observations and better estimation capabilities for deep-level information in the field. Figure 13 shows the layers in the maize canopy for the LAI observations.

To analyze how the model performs for the different maize genotypes, we applied a two-way ANOVA (Gómez-Gálvez et al., 2021). Three maize varieties, Fengken-139 (FK-139), Jingnongke-728 (JNK-728), and Zhengdan-958 (ZD-958), were used to determine whether significant differences occur in the estimated LAIs. IBM SPSS software (IBM Corp, Armonk, NY, USA) was used to analyze the variance tests. Figure 14 shows the LAI determined by field observations. The sowing time for the nitrogen experiment is June 14. ZD-958 produces a greater difference in LAI than the other

Table 8 Details of nitrogen-gradient management schemes set up in experimental field

Nitrogen Treatment No.1 (kg/hm^2)	N0	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10
Base Fertilizer	0	150	150	150	150	150	150	150	0	0	0
Leaf Stage 5–6	0	0	100	0	0	100	100	0	150	150	150
Leaf Stage 11–12	0	0	0	100	0	100	0	100	100	0	100
Silking Stage	0	0	0	0	100	0	100	100	0	100	100
Total	0	150	250	250	250	350	350	350	250	250	350

Nitrogen Treatment No.2 (kg/hm^2)	N0	N100	N200	N300	N400
Base Fertilizer	0	60	120	180	240
Leaf Stage 11–12	0	40	80	120	160
Total	0	100	200	300	400

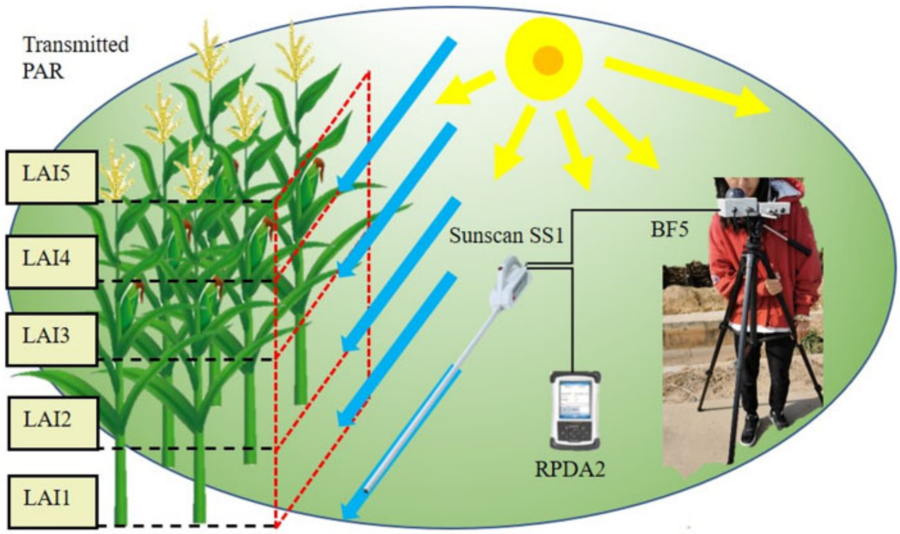

Figure 13 Field observation of maize LAI by Sunscan canopy analyzer. Note: The black and red dashed lines represent the height of LAI observation.

two germplasms. The same phenomenon occurs for the nitrogen-gradient grades from N100 to N400.

Due to the COVID-19 pandemic, we could not collect all UAV field images, which results in a lack of images of the early phenological traits of maize. The LAI of the early sown maize is generally low because the early sown maize germplasm material enters early into the physiologically mature stage, so the leaves have yellowed and withered. The 636 UAV images of maize plots were obtained during the entire maize growth period. Because some of the images could not be collected during the early stage of maize phenology (maize seedling), a total of 451 LAI observations were obtained.

UAV image acquisition

Figure 4 shows the UAV crop phenotype observation platform designed for this study, and Table 9 lists the sensor parameters.

This study used a hexacopter DJI Matrice M600pro (SZ DJI Technology Co., Ltd. China) electric UAV with a wheelbase of 1,133 mm as a low-altitude remote-sensing observation platform (see Figure 15A). The DJI Matrice M600pro is equipped with a professional DJI A3 flight control system, including a main controller, GPS-Compass Pro, Power Management Unit, and LED, and uses a Lightbridge 2 high-definition digital image transmission system. The digital camera and the MS camera were mounted on the UAV aircraft and used in synchronized shooting mode. To avoid field observation errors caused by cloud cover, the UAV flight experiment was done under clear and cloudless weather conditions. The acquisition time was 10:00–14:00, the flight altitude was 30 m, the heading overlap was 90%, and the side overlap was 80%. With zero wind, the flight speed was 1 m/s. To study how the development of the different maize genotypes depends on the environmental conditions, images were collected in each key growth period: tasseling (July 20), silking (August 1), blister (August 8), milk (August 16), dough (August 23), and dent (September 3). To calibrate the ASD FieldSpec FR 2500 hyperspectrometer, six calibration carpets of different colors (see Figure 15B) were placed in the UAV flight belt and their reflectance was simultaneously measured. All sensors were set to capture one image per second. In addition, 10 ground control points were placed around the field plot; these remained fixed throughout the phenological period of maize. Finally,

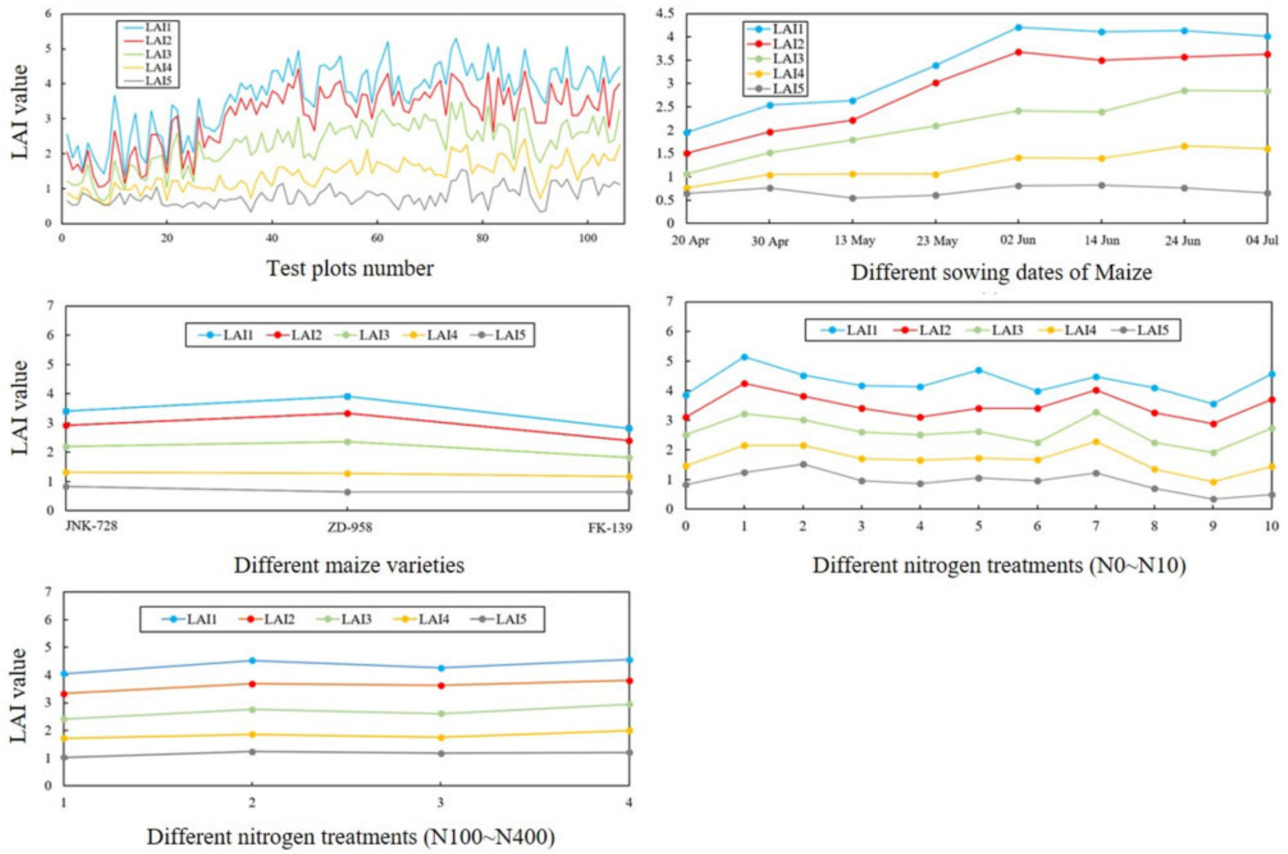

Figure 14 Results of field LAI observations under different experimental conditions and at different vertical levels of the maize plants. A, LAI observations of all germplasms, (B) LAI observations as a function of sowing date. C, LAI observations as function of maize variety. D, LAI observations as a function of nitrogen gradient (N0–N10). E, LAI observations as a function of nitrogen gradient (N100–N400).

Table 9 Sensor parameters for remote sensing sensors mounted on the UAV

Sensor	Sensor Type	Resolution (Pixels)	PW (μm)	FWHM (nm)	FOV(H° × V°)
SONY ILCE-7M2	RGB	6,000 × 4,000	—	—	—
MicaSense RedEdge-MX	MS	1,280 × 1,024	0.475, 0.560, 0.668, 0.717, and 0.840	20, 20, 10, 10, and 40	47.2° × 30.9°
FLIR DUO PRO R 640	RGB	4,000 × 3,000	—	—	56.0° × 45.0°
	Thermal	640 × 512	10.5	3,000	45.0° × 37.0°

PW, peak wavelength; FWHM, full width at half maximum; FOV, field of vision.

a high-precision GPS device served to obtain the center GPS coordinates of these reference points for later geometric correction and image registration.

The RGB camera (Figure 15D) was 126.9 × 95.7× 59.7 mm^3, 556 g SONY ILCE-7M2 (Sony Corporation, Tokyo, Japan), with 6,000 × 4,000 pixels, which corresponds to ∼24.3 million effective pixels. The camera recorded data in *.tif format and had an ISO range of shooting motion image sensitivity of 200–25,600. It was also equipped with a fast hybrid autofocus function and a five-axis anti-shake function to compensate for vibrations during image acquisition.

The Micasense RedEdge-MX (Micasense, Inc., Seattle, WA, USA) MS camera (Figure 15D) made MS image (1,280 × 960) per second (all frequency bands), recorded in 12-bit .tif, and weighed 231.9 g. It consisted of a RedEdge-MX + DLS 2, 8.7 × 5.9 × 4.54 cm sensor that captures in the blue, green, red, red edge, and near-infrared. The radiometric calibration of the UAV MS images was performed by imaging a color-calibration cloth imaged by the ground ASD hyperspectrometer to determine the band-response function in each band and thereby build an MS radiometric calibration model.

The 85 × 81.3 × 68.5 mm^3, 325 g TIR camera (Figure 15D) used a FLIR DUO PRO R 640 (FLIR Systems, Inc., Wilsonville, OR, USA) and was equipped with two lenses: a TIR lens (640 × 512) and an RGB lens (4,000 × 3,000). The temperature obtained by the ground-temperature measuring board (Figure 15C) served as the standard for calibrating the TIR images. While collecting TIR images from the UAV, the temperature sensor recorded the actual temperature of the black-board every 30 s. Temperature calibration equipment (black board) set up in the field is composed of a steel sheet with

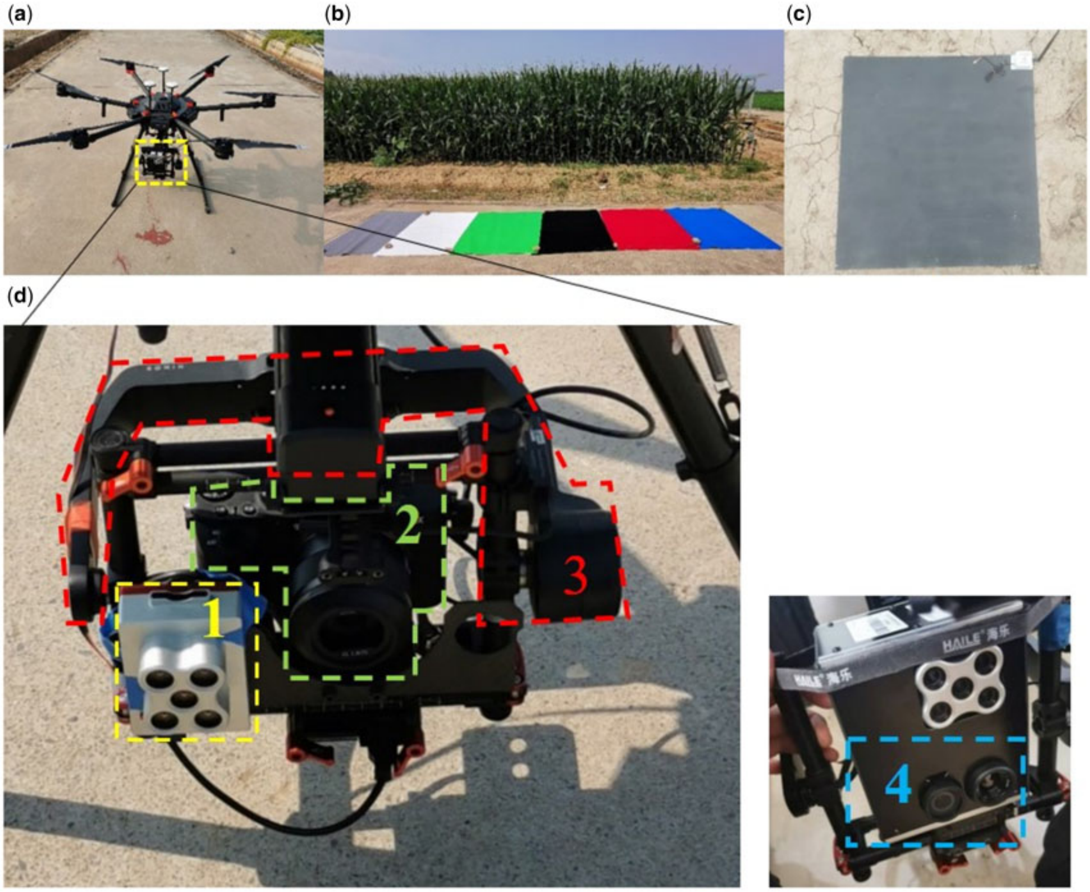

Figure 15 UAV flight system and integrated sensors. A, DJI Matrice 600 Hexacopter Platform. B, MicaSense RedEdge-MX MS imaging radiometric calibration carpet (blue, green, red, white, gray, and black). C, TIR black body. D, Multi-source remote sensing integrated equipment: 1, MS camera; 2, RGB camera; 3, Romin 2 cameras pan tilt; 4, TIR sensor.

blackbody paint coating and a temperature sensor. At noon, the sunlight is sufficient to make the surface temperature of the black board to a high value. Before stitching the UAV images, the DN of the blackboard was extracted by using the ENVI software, and a linear relationship was established between the actual temperature and the DN of the blackboard, which completed the calibration process of the TIR images (Maimaitijiang et al., 2020). The radiance was converted into the field crop canopy temperature by using the TIR image radiometric calibration. Figure 16 shows the measured temperature of the blackboard as a function of DN of thermal images.

The Agisoft PhotoScan Software (Agisoft LLC, St Petersburg, Russia) was used to stitch the UAV multi-source remote-sensing images. The software can splice and geometrically correct images based on GPS coordinates or image textures of the ground aviation marking points and can also compensate for variations in UAV attitude and atmospheric refraction. The RGB, MS, and TIR images collected for this study were all preprocessed by the PhotoScan software to produce the UAV orthophoto image. Figure 17 presents a flow chart detailing the overall procedure used in this study.

The maize plot sample data used in this study are cut from the UAV orthophoto images by using the data frame

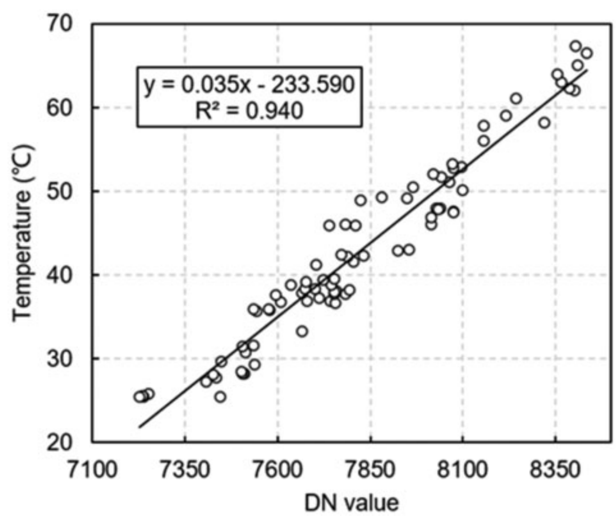

Figure 16 Measured temperature of blackboards plotted versus DN of thermal images. Solid line shows the linear fit given in the legend. Note: Digital Number (DN) is the luminance value of the image pixel.

shown in Figure 17. The .shp file is constructed by ArcGIS version 10.4, and the orthophoto image is cut to get the plot sample image in .tif format for follow-up analysis. The

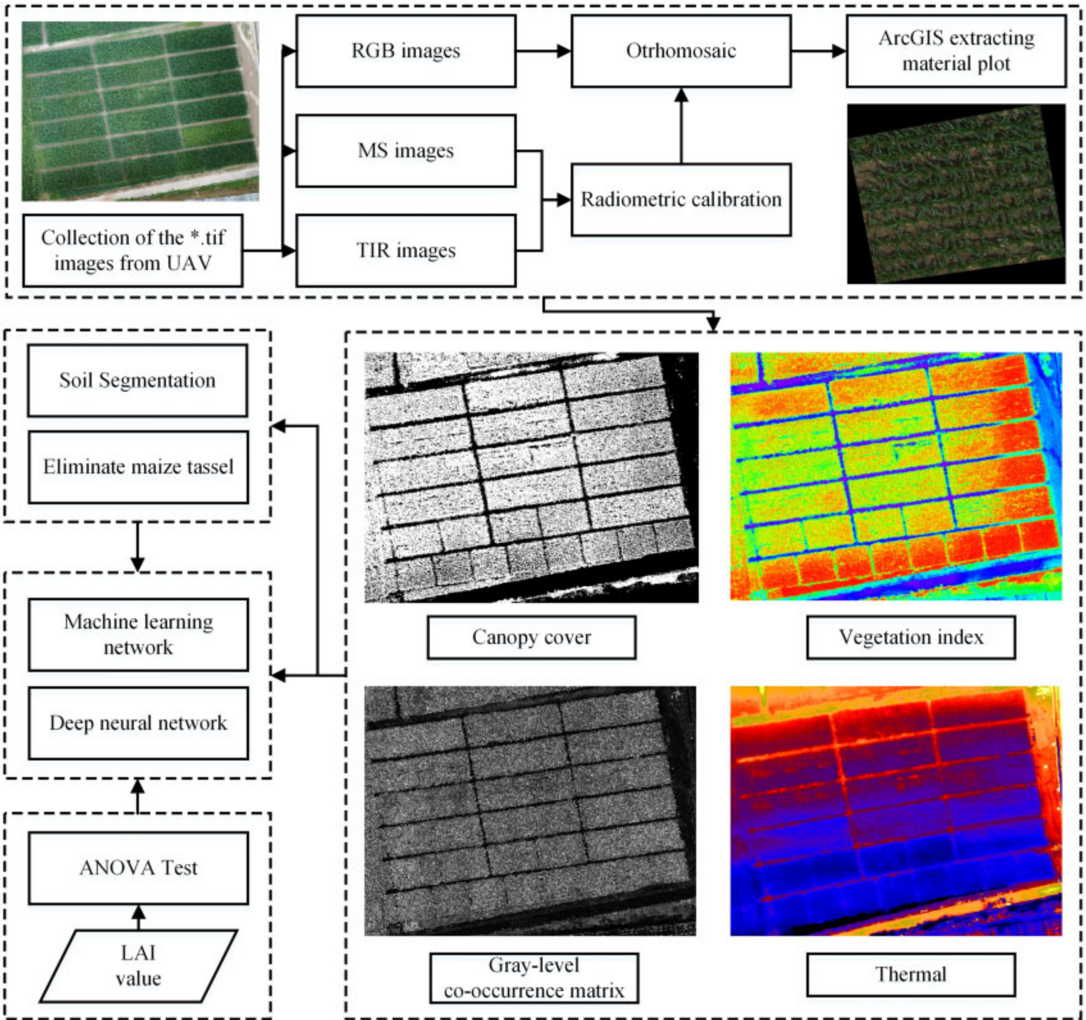

Figure 17 Multi-source remote sensing data fusion processing framework: Note: The color in the picture, CC: black represents the soil background, white represents the maize plant, vegetation index: warm colors represent high VI values, cool colors represent low VI values, GLCM: black represents the soil background, gray represents texture information, thermal: warm colors represent high-temperature regions, while cool colors represent low-temperature regions.

maize plot is cropped as a whole field with a single treatment condition and measurement (Figure 17). An experimental processing corresponds to a maize sample plot, and there are no controlled trials. After cropping the UAV orthophoto, ArcGIS version 10.4 and Python-GDAL version 2.2.4 were used to locate areas for subsequent research.

Method

In this study, quantitative information is extracted from UAV multimodal data and integrated into a 1-D vector. Compared with learning LAI features directly from the image, as done with a convolution neural network, the integration of multimodal data into 1-D vectors can greatly improve the computational efficiency, and the maize LAI can be estimated without a large number of samples. On this basis, the machine learning model and DNN are constructed to train 1-D vectors to estimate the maize LAI. This section introduces the construction of this 1-D vector and the procedure for estimating the maize LAI.

Extraction of canopy information

In this study, the vegetation index constitutes a spectral feature of the maize canopy and is based on the visible, red edge, and near-infrared bands of the UAV images. It was determined as per previous research and was used to estimate the LAI. With traditional satellite visible images, the long interval between the scanning bands of the satellite means that the optical elements are easily affected by the environment (e.g. atmosphere and topography). This leads to a certain deviation between the measured value obtained by the remote sensor and the reflected energy signal of the actual target (Teillet et al., 2007). Therefore, radiometric correction is often needed before using the remote sensing satellite images. With low-altitude shooting, most deviations caused by natural environmental factors are negligible; in particular, the influence of atmospheric factors can be ignored.

The RGB image can be collected by UAV in a short time at noon, which eliminates some of the errors caused by variations in light intensity. Compared with MS images, RGB images are not sensitive to the change of light intensity and emphasize more the reflection of texture information. The visible light vegetation index constructed by RGB image DN value provides reliable accuracy to estimate vegetation physical structure parameters (Li et al., 2020). Therefore, this article uses the DN value of the RGB image to construct a visible band vegetation index for estimating the maize LAI. Tables 10 and 11 list its sources and the specific vegetation index formula.

Canopy structure information

Vegetation CC, which is the land area per vegetation area in percent, was extracted as a variable to reflect the vegetation density and structure and was used to estimate the LAI (Schirrmann et al., 2016). The vegetation index reflects the vegetation growth status, physiological parameters, and coverage based on the relationship between the reflectance in the different bands and remains a primary method for extracting vegetation information (Makanza et al., 2018). The vegetation index in the visible (RGB) band was used to extract the CC. A binary mask map was first constructed to extract maize pixels from the UAV images using the vegetation indexes CIVE, EXG, EXGR, RGBVI, VDVI, and VEG. The constructed production mask then serves to divide the image into foreground (vegetation) and background (soil and other ground objects). Next, the maize pixels classified in each plot are divided by the total number of pixels in the plot to calculate the CC (Yu et al., 2016). The specific calculation is as follows:

$$P_{canopy} = \begin{cases} if\ a \geqslant P_{value} \geqslant b : P_{value} = 1 \\ ifP_{value} < borP_{value} > a : P_{value} = 0 \end{cases} \quad \text{(Eq. 1)}$$

$$\text{Canopy Cover}\ (CC) = \frac{\sum P_{canopy}}{\sum P_{value}} \times 100\% \quad \text{(Eq. 2)}$$

where P_{canopy} is the value of the pixel CC mask, $P_{value}$is the DN value of pixel, and CC is the coverage of the maize germplasm genotype, a and b are the empirical values used for threshold segmentation of the vegetation index.

UAV TIR images can provide the maize-canopy temperature information and can also give the information about maize LAI. Due to the difference in light interception, maize leaf temperature depends significantly on the height of the leaf on the maize crop. The normalized relative canopy temperature (Elsayed et al., 2015, 2017) was constructed from the calibrated TIR image from the UAV and then used to estimate the maize LAI.

We extracted texture features from the original bands obtained from the RGB, MS, and TIR sensors and used them as input image variables for machine learning models (Maimaitijiang et al., 2020). The GLCM (Haralick et al., 1973), which is a commonly used texture algorithm, served to test the capacity of UAV multi-source remote-sensing

Table 10 UAV RGB image variables and vegetation index formula

Spectral Indices	Definition	References
Red	$Rawvalueofredband(\boldsymbol{R})$	—
Green	$Rawvalueofgreenband(\boldsymbol{G})$	—
Blue	$Rawvalueofblueband(\boldsymbol{B})$	—
Normalized Red	$\boldsymbol{r} = \boldsymbol{R}/(\boldsymbol{R}+\boldsymbol{G}+\boldsymbol{B})$	—
Normalized Green	$\boldsymbol{g} = \boldsymbol{G}/(\boldsymbol{R}+\boldsymbol{G}+\boldsymbol{B})$	—
Normalized Blue	$\boldsymbol{b} = \boldsymbol{B}/(\boldsymbol{R}+\boldsymbol{G}+\boldsymbol{B})$	—
Green Red Ratio Index	$\text{GRRI} = \boldsymbol{G}/\boldsymbol{R}$	—
Green Blue Ratio Index	$\text{GBRI} = \boldsymbol{G}/\boldsymbol{B}$	—
Red Blue Ratio Index	$\text{RBRI} = \boldsymbol{R}/\boldsymbol{B}$	—
Color Intensity Index	$\text{INT} = (\boldsymbol{R}+\boldsymbol{B}+\boldsymbol{G})/3$	Ahmad and Reid, (1996)
Green–Red Vegetation Index	$\text{GRVI} = (\boldsymbol{G}-\boldsymbol{R})/(\boldsymbol{G}+\boldsymbol{R})$	Tucker (1979)
Normalized Difference Index	$\text{NDI} = (\boldsymbol{r}-\boldsymbol{g})/(\boldsymbol{r}+\boldsymbol{g}+0.01)$	Woebbecke et al. (1993)
Woebbecke Index	$\text{WI} = (\boldsymbol{G}-\boldsymbol{B})/(\boldsymbol{G}+\boldsymbol{R})$	Woebbecke et al. (1995)
Kawashima Index	$\text{IKAW} = (\boldsymbol{R}-\boldsymbol{B})/(\boldsymbol{R}+\boldsymbol{B})$	Kawashima and Nakatani (1998)
Green Leaf Index	$\text{GLI} = (2\times\boldsymbol{G}-\boldsymbol{R}-\boldsymbol{B})/(2\times\boldsymbol{G}+\boldsymbol{R}+\boldsymbol{B})$	Louhaichi et al. (2001)
Green Leaf Index 2	$\text{GLI2} = (2\times\boldsymbol{G}-\boldsymbol{R}+\boldsymbol{B})/(2\times\boldsymbol{G}+\boldsymbol{R}+\boldsymbol{B})$	Louhaichi et al. (2001)
VARI	$\text{VARI} = (\boldsymbol{G}-\boldsymbol{R})/(\boldsymbol{G}+\boldsymbol{R}-\boldsymbol{B})$	Gitelson et al. (2002)
Excess Red Vegetation Index	$\text{ExR} = 1.4\times\boldsymbol{r}-\boldsymbol{g}$	Mao et al. (2003)
Excess Green Vegetation Index	$\text{ExG} = 2\times\boldsymbol{g}-\boldsymbol{r}-\boldsymbol{b}$	Mao et al. (2003)
Excess Blue Vegetation Index	$\text{ExB} = 1.4\times\boldsymbol{b}-\boldsymbol{g}$	Mao et al. (2003)
Excess Green Minus Excess Red Index	$\text{ExGR} = \text{ExG}-\text{ExR}$	Mao et al. (2003)
Vegetative Index	$\text{VEG} = \boldsymbol{G}/(\boldsymbol{R}^{(2/3)}\times\boldsymbol{b}^{(1/3)})$	Hague et al. (2006)
Principal Component Analysis Index	$\text{IPCA} = 0.994\times\boldsymbol{R}-\boldsymbol{B}+0.961\times\boldsymbol{G}-\boldsymbol{B}+0.914\times\boldsymbol{G}-\boldsymbol{R}$	Saberioon et al. (2014)
Color Index of Vegetation	$\text{CIVE} = 0.441\times\boldsymbol{r}0.811\times\boldsymbol{g}+0.3856\times\boldsymbol{b}+18.78745$	Kataoka et al. (2003)
Combination	$\text{COM} = 0.25\times\text{ExG}+0.3\times\text{ExGR}+0.33\times\text{CIVE}+0.12\times\text{VEG}$	Guijarro et al. (2011)
Red Green Blue Vegetation Index	$\text{RGBVI} = (\boldsymbol{G}^2-\boldsymbol{B}\times\boldsymbol{R})/(\boldsymbol{G}^2+\boldsymbol{B}\times\boldsymbol{R})$	Gamon and Surfus (1999)
Modified Green Red Vegetation Index	$\text{MGRVI} = (\boldsymbol{G}^2-\boldsymbol{R}^2)/(\boldsymbol{G}^2+\boldsymbol{R}^2)$	Bendig et al. (2015)

Note: R, G, and B were collected from UAV images, which mean the DN value of the maize pixel.

Table 11 Spectral indices of UAV MS image

Spectral Indices	Definition	References
Difference Vegetation Index	$\mathrm{DVI} = R_{nir} - R_{red}$	Tucker (1979)
Enhanced Vegetation Index	$\mathrm{EVI} = 2.5 \times (R_{nir} - R_{red})/(R_{nir} + 6 \times R_{red} - 7.5 \times R_{blue} + 1)$	Huete et al. (2002)
Green Normalized Difference Vegetation	$\mathrm{GNDVI} = (R_{nir} - R_{green})/(R_{nir} + R_{green})$	(Gitelson et al., 1996)
Ratio Between NIR and Green Bands	$\mathrm{VI}_{(nir/green)} = R_{nir}/R_{green}$	Gitelson and Merzlyak (1997)
Ratio Between NIR and Red Bands	$\mathrm{VI}_{(nir/red)} = R_{nir}/R_{red}$	Ramoelo et al. (2012)
Ratio Between NIR and Red Edge Bands	$\mathrm{VI}_{(nir/rededge)} = R_{nir}/R_{green}$	Ramoelo et al. (2012)
Neperian Logarithm of The Red Edge	$\mathrm{In}_{RE} = 100 \times (\mathrm{In}_{nir} - \mathrm{In}_{red})$	Portz et al. (2012)
Modified Soil Adjusted Vegetation Index	$\mathrm{MSAVI} = (1 + L)\left(\frac{R_{nir} - R_{red}}{R_{nir} + R_{red} + L}\right)(L = 0.1)$	Qi et al. (1994)
Modified SAVI 2	$\mathrm{MSAVI2} = R_{nir} + 0.5 - \sqrt{(2 \times R_{nir} + 1)^2 - 8 \times (R_{nir} - R_{red})/2}$	Qi et al. (1994)
Optimized SAVI	$\mathrm{OSAVI} = (1 + 0.16) \times \frac{(R_{nir} - R_{red})}{(R_{nir} + R_{red} + 0.16)}$	Geneviève et al. (1996)
Modified Triangular Vegetation Index 2	$\mathrm{MTVI2} = \frac{1.5 \times [1.2 \times (R_{nir} - R_{green}) - 2.5 \times (R_{red} - R_{green})]}{\sqrt{(2 \times R_{nir} + 1)^2 - \left(6 \times R_{nir} - 5 \times \sqrt{R_{red}}\right) - 0.5}}$	Haboudane et al. (2004)
Normalized Difference Red Edge Index	$\mathrm{NDRE} = \frac{(R_{nir} - R_{rededge})}{(R_{nir} + R_{rededge})}$	Gitelson and Merzlyak (1994)
Normalized Difference Vegetation Index	$\mathrm{NDVI} = \frac{(R_{nir} - R_{red})}{(R_{nir} + R_{red})}$	Myneni and Williams (1994)
Modified Simple Radio	$\mathrm{MSR} = (R_{nir} - R_{red} - 1)/\left(\sqrt{R_{nir} + R_{red}} + 1\right)$	Chen (1996)
SAVI	$\mathrm{SAVI} = \frac{(R_{nir} - R_{red})}{(R_{nir} + R_{red} + 0.5)} \times (1 + 0.5)$	Huete (1988)
Simplified Canopy Chlorophyll Content Index	$\mathrm{SCCCI} = \frac{\mathrm{NDRE}}{\mathrm{NDVI}}$	Raper and Varco (2015)
Modified Chlorophyll Absorption Reflectance Index	$\mathrm{MCARI} = (R_{rededge} - R_{red} - 0.2 \times (R_{rededge} - R_{green})) \times \left(\frac{R_{rededge}}{R_{red}}\right)$	Daughtry et al. (2000)
Modified Chlorophyll Absorption Reflectance Index 2	$\mathrm{MCARI2} = 1.5 \times \frac{(2.5 \times (R_{nir} - \mathrm{R}_{rededge}) - 1.3 \times (R_{nir} - R_g))}{\left(2 \times (R_{nir} + 1)^2 - \left(6 \times R_{nir} - 5 \times (R_{red})^2\right) - 0.5\right)}$	Haboudane et al. (2004)
Transformed Chlorophyll Absorption Reflectance Index	$\mathrm{TCARI} = 3 \times \left((R_{rededge} - R_{red}) - 0.2 \times (R_{rededge} - R_{green}) \times \left(\frac{R_{rededge}}{R_{red}}\right)\right)$	Haboudane et al. (2002)
Normalized Difference Index	$\mathrm{NDI} = \frac{(R_{nir} - R_{rededge})}{(R_{nir} + R_{red})}$	Datt (1999)
Red-Edge Chlorophyll Index 1	$\mathrm{CI1} = \frac{R_{nir}}{R_{rededge}} - 1$	Li et al. (2012)
Red-Edge Chlorophyll Index 1	$\mathrm{CI2} = \frac{R_{rededge}}{R_{green}} - 1$	Gitelson and Merzlyak (1996)
Structure-Insensitive Pigment Index	$\mathrm{SIPI} = \frac{(R_{nir} - R_{blue})}{(R_{nir} + R_{red})}$	Penuelas et al. (1995)
TCARI/OSAVI	$\frac{\mathrm{TCARI}}{\mathrm{OSAVI}}$	Haboudane et al. (2002)
MCARI/OSAVI	$\frac{\mathrm{MCARI}}{\mathrm{OSAVI}}$	Zarco-Tejada et al. (2004)

Note: $R_{rededge}$ and R_{nir} are the red edge and near-infrared band reflectance values in the MS image.

data to estimate the maize LAI. The GLCM gives the GLCM-based texture information, which includes the contrast, correlation, dissimilarity, entropy, homogeneity, mean, second moment, and variance (Nichol and Sarker, 2011). This study used the ENVI version 5.4 (Exelis Visual Information Solutions, Boulder, CO, USA) data processing software to extract the GLCM texture information.

DNN model

Multimodal data fusion can use data from different sensors to obtain different types of feature information and extract the effective features, which can improve the learning efficiency. The framework of DNNs allows for feature-level fusion of multi-source data (Williams et al., 2018). The input-level feature fusion method integrates data from different data sources and estimates the LAI of maize germplasm with the goal of constructing a fully connected feedforward neural network (Biganzoli et al., 1998). The fully connected feedforward DNN increases the number of layers of the neural network. The goal of stacking multiple layers is to extract higher-order feature information, which can be beneficial for performance in some scenarios (Langkvist et al., 2014). It can generate adaptive, robust relationships and trained models that can be applied very quickly. As a result, the DNN model has become a machine learning solution for various classification and regression problems (Cai et al., 2018; Zhang et al., 2018). Although DL models are peculiarly prone to overfitting when not appropriately regularized, they are effective for identifying the primacy of different independent variables and thus more accurately estimating crop phenotyping indicators (Chlingaryan et al., 2018). For this work, we designed the input-level feature multimodal DNN structure depicted in Figure 18.

The DNN model in this article is more like a multilayer perceptual (MLP) machine model. MLP-DNN models have a hidden layer, with the vector being the only input. MLP is suitable for classified predictions and regression forecast issues. When the CNN convolutionary is the same size as the input data, its calculation is equivalent to that done by the MLP model (e.g. the input picture is 100 × 100, the volume subscmoid size is 100 × 100), which directly results in a loss of input spatial information. Therefore, this model is unsuited for images with spatial information. Maimaitijiang et al. (2020) built this network structure and successfully predicted the soybean (*Glycine max*) yield. By quantitative processing, the feature information was extracted from the image and combined into a 1-D vector. The quantitative information extracted from each maize plot is integrated into a 1-D vector and then entered into the DNN model for training.

We designed a fully connected input layer and a hidden layer and connected them to the final regression on the final fully connected layer to estimate the LAI. The DNN was set up in five layers: one input layer, three hidden layers, and one output layer. The texture, thermal information, and vegetation index extracted from the image were compressed into a 1-D vector corresponding to each maize germplasm plot and served as the input layer for the model. The hidden layer contained three convolution layers, with each hidden layer having 10 neurons. In the DNN model, the selected activation function was $f = \max(0, z)$ with rectified linear units. The estimated LAI was output linearly from the last layer of the DNN. Keras provides a variety of loss functions, such as mean squared error, mean absolute percentage error, mean absolute error, mean squared logarithmic error, squared hinge, logcosh, categorical crossentropy, and binary crossentropy. Herein we use the mean squared error to calculate model loss. This formula is given as:

$$L = \frac{1}{n}\sum\nolimits_{i=1}^{n}\left(y_{\text{true}}^{(i)} - y_{\text{pred}}^{(i)}\right)^2 + \frac{\lambda}{2n}||w||_2^2 \qquad \text{(Eq. 3)}$$

$$||w||_2^2 = \sum\nolimits_{j=1}^{n} w_j^2 = w^T w \qquad \text{(Eq. 4)}$$

where n is the total number of samples, $y_{\text{ture}}^{(i)}$ is the LAI observations of the maize plot, $y_{\text{pred}}^{(i)}$ is the LAI estimated by the DNN model, λ is the regularization parameter, and w is the weight of each neuron.

Lack of sufficient training data or overtraining will lead to overfitting by the model. Regularization can reduce the complexity and instability of the model in the learning process, thereby avoiding overfitting. To solve the problem of overfitting, we used an L2 regularization to prevent overfitting of the model (see Equations 2–4).

The initial weight of the model was randomly assigned and then updated by using the gradient descent algorithm of the Adam optimizer. Each parameter in the DL model is solved for its gradient value. Under a certain learning rate, the parameters are updated in the opposite direction of the gradient. The Adam optimizer dynamically modifies the learning rate of each parameter. In addition, the momentum method was added, which increases the efficiency of the parameter update to optimize the local result. The DL model epochs were set to 10 000. The average time for each iteration was 30 ms/step. The computer used in this article was equipped with NVIDIA 960M graphics card, an Intel Core i7-6700HQ CPU, and 16 GB of memory. The computer compiler was running Python version 3.6.1 and Keras version 2.4.3. Table 12 gives the specific properties and numerical settings of the DNN model.

PLSR

PLSR is a multivariate statistical method that is widely used in spectroscopy and regression analysis to analyze the correlation between multiple predictor variables and corresponding variables (Duan et al., 2019). The optimal number

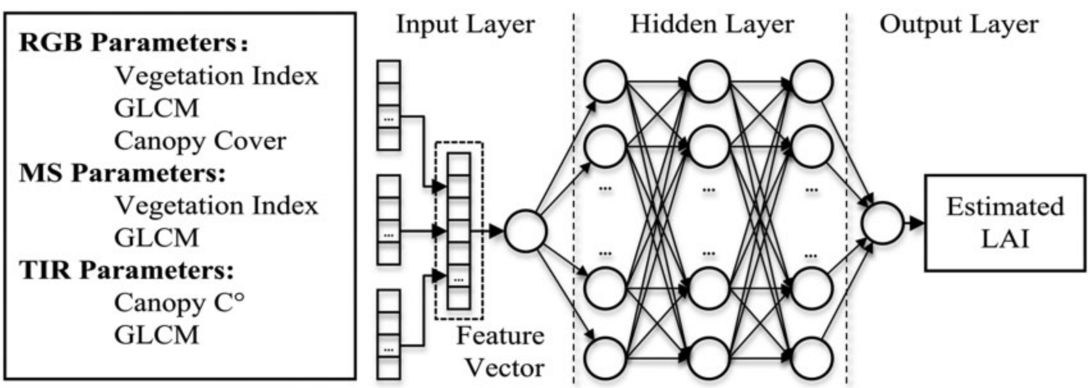

Figure 18 Schematic illustration of multimodal feature input level deep network structure used herein to estimate the LAI. Note: the circles represent the neural unit; arrows represent the output direction of the input data; the three dots represent neurons that cannot be fully displayed; the dotted frame in the picture represents "Feature Vector."

Table 12 Parameters used in DNN estimation model

Model Parameters	Value
Normalize	L2
Optimizer	Adam (Adaptive moment estimation)
The Activation Function	Rectified linear unit (hidden layer), linear (output layer)
Batch Size	64
Learning Rate	0.01
Epochs	10,000
Verbose	1
Dropout	0.5

of potential components in PLSR is determined by minimizing the RMSE of leave-one-out cross-validation. Compared with MLR models, PLSR is characterized by the ability to perform regression modeling even when the independent variables have severe multiple correlations. PLSR allows regression modeling even when given fewer sample points than variables, and all original independent variables are included in the final model. We constructed in Python a Scikit-learn version 0.19.1 Cross_decomposition PLSR estimation model. Table 13 gives the specific properties and numerical settings of the model.

Support vector machine regression

SVR is a widely used machine learning method based on the key concepts of statistical learning theory (Song et al., 2020). SVR transforms the input sample set into a high-dimensional space to improve the separation of samples. Its structure is like a three-layer perceptron, which is a general method for constructing classification (Chen et al., 2020). When applying a SVM to the regression problem, a boundary is defined, and the points within the boundary are considered to have zero error. SVMs are efficient in finding sample information to achieve the best compromise between model complexity and learning ability. An SVM can linearly separate highly nonlinear data by using a kernel function. An SVM was constructed by using Python's Scikit-learn version 0.19.1-SVM version 0.19.1. Table 14 lists the specific properties and numerical settings of the model.

RFR

RFR is a nonlinear statistical ensemble method that can construct and then average a large number of random and uncorrelated decision trees for classification or regression (Hastie et al., 2001). For regression modeling tasks, the main advantage is that the risk of overfitting can be

Table 13 Parameters used in PLSR estimation model

Model Parameters	Value
N_components (Number of Components to Keep)	4.0
Max_iter (The Maximum Number of Iterations)	500
Tol (Tolerance Used in the Iterative Algorithm)	$1e^{-06}$
Scale (Scale the Data)	True

Table 14 Parameters used in SVM regression estimation model

Model Parameters	Value
C (Regularization Parameter)	1.0
Kernel (Specifies the Kernel Type)	Linear
Max_iter	−1 (no limit)
Tol (Tolerance Used in the Iterative Algorithm)	10^{-3}
Cache_size	200

minimized. At the same time, variable importance scores can also be calculated to evaluate the contribution to the model of each predictor variable, thereby eliminating redundant variables (Christopher and Michael, 2016). The user need only specify the parameters of the model and select the number of variables at each split and the total number of trees to grow (Christopher and Michael, 2016). To find the optimal value of these two parameters, the total number of best-grown trees was tested from 100 to 1,000 in intervals of 100. The tree sum that produced the lowest RMSE was then selected as the parameters of the best model. The RFR model uses the Scikit-learn version 0.19.1 library and was built in Python version 3.6.1 (Python Software Foundation, Wilmington, DE, USA). RFR was constructed by using Python's Sklearn version 0.19.1-RandomForestRegressor. Table 15 lists the specific properties and numerical settings of the model.

Model evaluation

The amount of data available for each growth period is shown in Table 1. The cut UAV image maize plot and LAI observations were taken as a group of samples. When verifying the estimation effect of LAI model during the whole growth period, we randomly selected 66.7% of the samples (301 maize plots) and the observed LAI to train the model, with the remaining 33.3% of the samples (150 maize plots) used to test the performance and accuracy of the estimation model. When we used the machine learning model to estimate the LAI without regularized data, the model remained accurate. Data regularization does not solve the problem of linear correlation between features in the data. Moreover, bias is added to reduce the variance. Although the machine learning model does not use normalized data, the accuracy of the results is basically unaffected, and the two are basically the same. Therefore, the original feature information was used to train the machine learning models. The R^2, RMSE, and rRMSE of each model were calculated to evaluate the performance and accuracy of the LAI estimation model. The specific calculations are given below:

Table 15 Setting of parameters in RFR estimation model

Model Parameters	Value
n_estimators (number of trees in the forest)	20
n_jobs	−1
Random_state	50
Min_samples_leaf	1
Min_impurity_split	10^{-7}

$$R^2 = \sum_{i=1}^{n} (\hat{y}_i - \bar{y}_i)(y_i - \bar{y}) / \sqrt{\sum_{i=1}^{n} (\hat{y}_i - \bar{y}_i)^2 \sum_{i=1}^{n} (y_i - \bar{y})^2} \quad \text{(Eq. 5)}$$

$$\text{RMSE} = \sqrt{\sum_{i=1}^{n} (\hat{y}_i - y_i)^2 / n} \quad \text{(Eq. 6)}$$

$$\text{MAE} = \sum_{i=1}^{n} \hat{y}_i - y_i / n \quad \text{(Eq. 7)}$$

$$\text{rRMSE} = \left(\sqrt{\sum_{i=1}^{n} (\hat{y}_i - y_i)^2 / n} \right) / \bar{y} \quad \text{(Eq. 8)}$$

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